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1 Intro to data

Exercise 1.1. Within the framework of the Munich Rent Report, several variables were surveyed. Discuss the variables `net rent` (in Euro), `living size` (in m²), `building age` (in years), `central heating` (yes/no), `hot water supply` (yes/no), `good location` (yes/no), and `upscale bathroom equipment` (yes/no) regarding their variable type (i.e., nominal, ordinal, numeric).

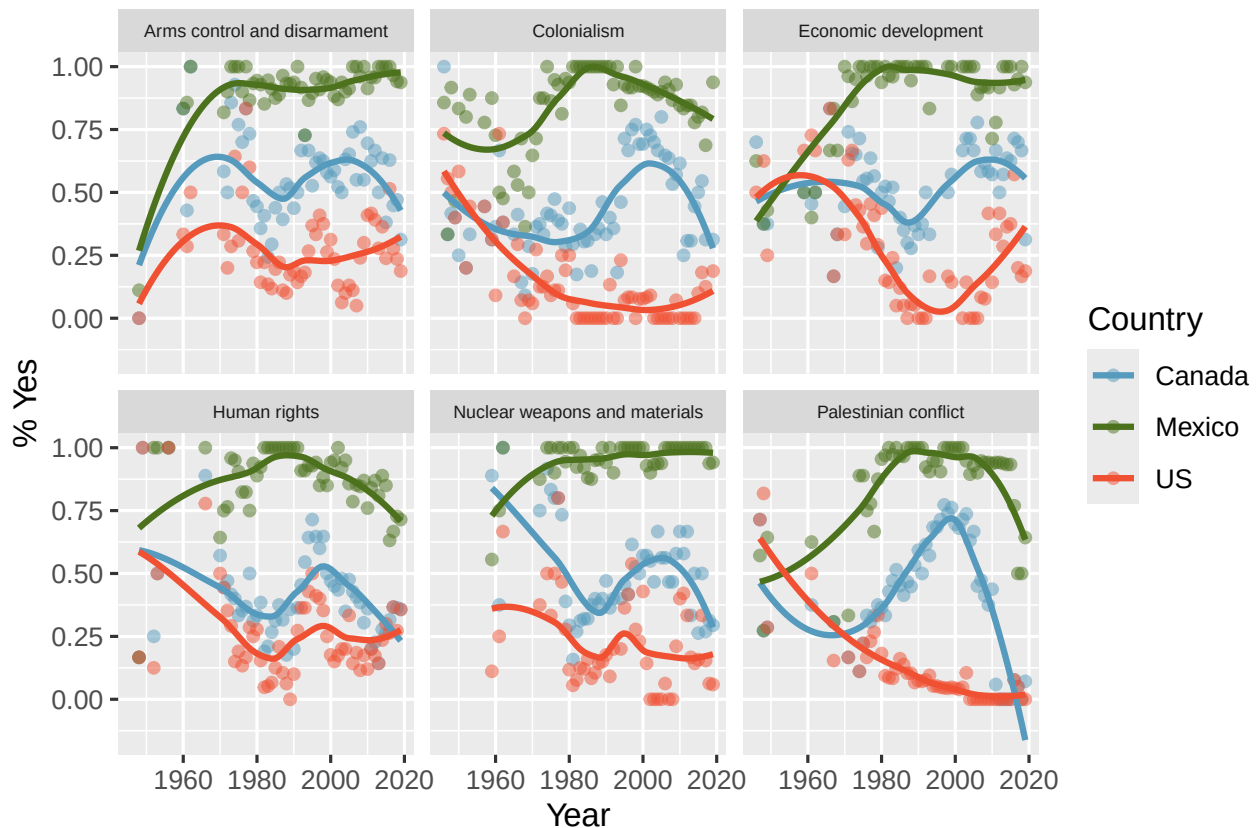
Solution:

The following variables should be discussed: `net rent`, `living size`, `building age`, `central heating`, `hot water supply`, `good location`, and `upscale bathroom equipment`.

The classification of the variables is as follows:

- `central heating` (nominal)
- `good location` (ordinal)
- `net rent` (numeric)
- `hot water supply` (nominal)
- `upscale bathroom equipment` (ordinal)
- `living size` (numeric)
- `building age` (numeric)

Exercise 1.2. The visualization below shows voting patterns on various issues in the United States, Canada, and Mexico in the United Nations General Assembly. Precisely, for a given year between 1946 and 2015, it displays the percentage of roll calls in which the country voted yes for each issue. This visualization was constructed based on a dataset where each observation is a country/year pair.



- List the variables used in creating this visualization.
- Indicate whether each variable in the study is numerical or categorical. If categorical, indicate if the variable is ordinal.

Solution:

- The variables used in creating this visualization are:

- Country
- Year
- Issue
- Percentage of Yes votes

- The variables have the following scales:

- Country: categorical
- Year: numerical
- Issue: categorical
- Percentage of Yes votes: numerical

Both categorical variables are nominal. The Country values cannot be ordered, and there is no universally accepted natural ordering for Issues.

Exercise 1.3. A statistical consulting company is charged with the following topics:

- a) Quality tests of wine from Koblenz.
- b) Quality inspections of condom production.
- c) Analyzing the addiction behavior of youths.

As head of the data collection department, you should decide between a census (full survey of the whole population) and a sample survey to collect the required data. Give reasons for your decisions.

Solution:

- a) If testing wine means tasting wine from a bottle, you can only use a random sample. This is because after opening the bottle and drinking (part of) the wine, the bottle can't be sold anymore. Hence, we can't do this for all the produced bottles.
If we could test the wine in the winery before the wine is filled into bottles and if we could additionally claim that tasting some part of one charge of wine is representative for all the bottles that will be filled from that charge, we might say, that we do a full survey.
- b) To guarantee quality, a full survey is essential. But this can be done in different ways. One test procedure, e.g., can be applied to each condom. In this test procedure, an appropriate machine is used, which allows the testing of condoms without harming them. Another test procedure would consider just a sample from a pre-specified number of condoms (often about 150000). If the failure rate in the sample is too high, all the 150000 condoms are thrown away.
- c) A census would be preferable here. However, because of cost reasons and lacking information (who is actually addicted?), the survey has to be limited to a sample.

Exercise 1.4. To improve the planning of parking spaces and the use of company-owned buses, a company is interested in some information about its employees. Specifically, the company is interested in the distance from their place of residence and the means of transport predominantly used by their employees to reach the workplace. To this end, the company uses a survey in which the economic situation of the employees was assessed. On a reference date, 50 employees were selected and questioned on the following aspects:

- household size (number of people living in the household),
- monthly rent,
- means of transport predominantly used to reach the workplace,
- distance between the place of residence and the place of work,
- own assessment of economic situation with 1 = very good, ... , 5 = very bad.

a) Specify the population and the observational units (the cases).

b) Which values can the collected variables take, and of which type are these variables?

Solution:

a) All employees of the company form the basic population. The 50 selected employees are the observational units.

b) We summarize the solution in the following way: variable name (possible values, type)

- **household size** ($\mathbb{N}$, numeric)
- **monthly rent** ($\mathbb{R}_0^+$, numeric)
- **means of transport** ({car, bus, bike, ...}, nominal)
- **distance** ($\mathbb{R}^+$ or $\mathbb{R}_0^+$, numeric)
- **assessment of situation** ({1,2,3,4,5}, ordinal)

Exercise 1.5. A study that surveyed a random sample of otherwise healthy high school students found that they are more likely to get muscle cramps when they are stressed. The study also noted that students drink more coffee and sleep less when they are stressed.

- a) What type of study is this?
- b) Can this study be used to conclude a causal relationship between increased stress and muscle cramps?
- c) State possible confounding variables that might explain the observed relationship between increased stress and muscle cramps.

Solution:

- a) Observational.
- b) Since the study is observational, we cannot infer causation.
- c) Caffeine and lack of sleep are potentially confounding variables that might explain the observed relationship between stress and muscle cramps, i.e., students might be experiencing muscle cramps due to increased caffeine consumption or lack of sleep.

Exercise 1.6. A researcher is interested in the effects of exercise on mental health and he proposes the following study:

- Use stratified random sampling to ensure representative proportions of individuals aged 18-30, 31-40, and 41-55 from the population.
 - Next, randomly assign half the subjects from each age group to exercise twice a week, and instruct the rest not to exercise.
 - Conduct a mental health exam at the beginning and at the end of the study, and compare the results.
- a) What type of study is this?
- b) What are the treatment and control groups in this study?
- c) Does this study make use of blocking? If so, what is the blocking variable?
- d) Does this study make use of blinding?
- e) Comment on whether or not the study results can be used to establish a causal relationship between exercise and mental health, and indicate whether or not the conclusions can be generalized to the population at large.
- f) Suppose you are tasked with determining if this proposed study should get funding. Would you have any reservations about the study proposal?

Solution:

- a) This is an experiment.
- b) The treatment group has to exercise twice a week, and the control group does not do any exercise.
- c) Yes, the blocking variable is age.
- d) No, the study is not blinded since the patients will know whether or not they are exercising.
- e) Since this is an experiment, we can make a causal statement. Since the sample is random, the causal statement can be generalized to the population at large. However, we should be cautious about making a causal statement because of a possible placebo effect.
- f) It would be challenging, if not impossible, to successfully conduct this study since randomly sampled people cannot be required to participate in a clinical trial.

2 Data vizualization / wrangling

Remark: Use the Quarto file `starwars-template.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 2.1. In this exercise, we will use the `starwars` dataset, which is included in the `dplyr` package. Because `ggplot2` is also required, we will load the `tidyverse` package.

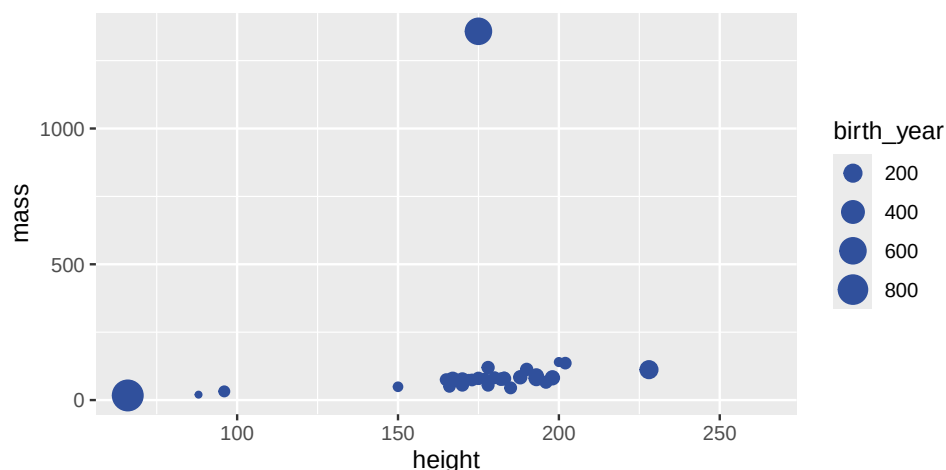
```
library(tidyverse)
```

Begin by examining the data. The `starwars` object is a tibble, so running the following code provides a concise overview of its structure.

```
starwars
# # A tibble: 87 x 14
#   name      height  mass hair_color skin_color eye_color birth_year sex  gender
#   <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
# 1 Luke Sk~    172    77 blond      fair        blue         19  male  mascu~
# 2 C-3P0      167    75 <NA>       gold        yellow       112  none  mascu~
# 3 R2-D2       96    32 <NA>       white, bl~  red          33  none  mascu~
# 4 Darth V~   202   136 none       white       yellow       41.9  male  mascu~
# 5 Leia Or~   150    49 brown      light       brown        19  fema~  femin~
# 6 Owen La~   178   120 brown, gr~ light       blue         52  male  mascu~
# 7 Beru Wh~   165    75 brown      light       blue         47  fema~  femin~
# 8 R5-D4       97    32 <NA>       white, red  red          NA  none  mascu~
# 9 Biggs D~   183    84 black      light       brown        24  male  mascu~
# 10 Obi-Wan~  182    77 auburn, w~ fair        blue-gray    57  male  mascu~
# # i 77 more rows
# # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
# #   vehicles <list>, starships <list>
```

a) Modify the following plot so that all points are displayed in pink.

```
ggplot(starwars,
       aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "#30509C")
```



b) Add labels for the plot title, x-axis, y-axis, and point size. Uncomment the relevant lines to observe the changes.


```
ggplot(starwars,
       aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "#30509C") +
  labs(
    #title = "___",
    #x = "___",
    #y = "___",
    #___
  )
```

- c) Select one categorical variable from the dataset and create a bar plot to display its distribution.

A bit of starter code is provided below, and the code chunk is set not to be evaluated with `eval: false` because the current code is not valid, so that the document would not render. Once you replace the code with valid code, set the chunk option to `eval: true` or remove the `eval` option altogether since it is set to `true` by default.

```
ggplot(starwars, aes(___)) +
  geom___
```

- d) Select one numerical variable and create a histogram to visualize its distribution.

This time, no starter code is provided. You are on your own!

```
# add code here
# pay attention to correctness and code style
```

- e) Select one numerical and one categorical variable, and create a visualization of their relationship using a method of your choice. Provide both your code and an interpretation of the resulting plot.

```
# add code here
# pay attention to correctness and code style
```

- f) Select two numerical and two categorical variables, and construct a visualization that incorporates all four variables. Include an interpretation with your response.

```
# add code here
# pay attention to correctness and code style
```

Solution:

We want to work with the `starwars` dataset, which is part of the `dplyr` package. Since we also need `ggplot2`, we load the `tidyverse`.

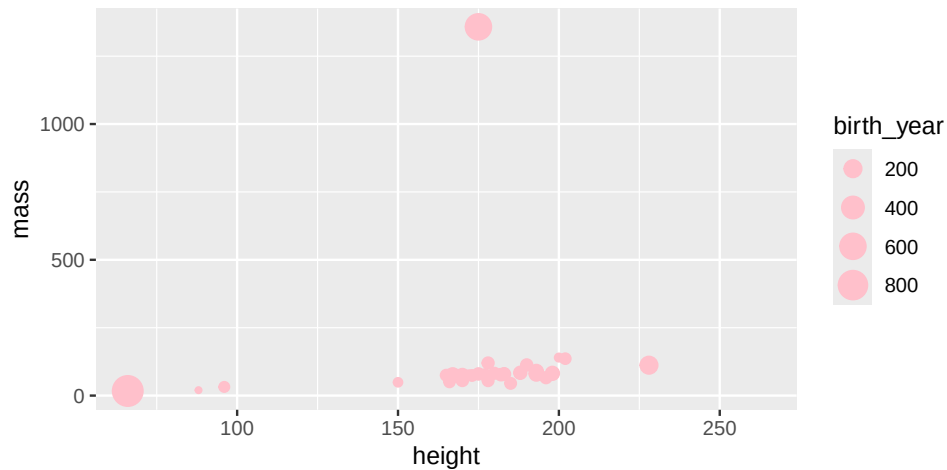
```
library(tidyverse)
```

Let's start by having a look at the data. Since the object `starwars` is a tibble, we get an excellent overview by just running the following code:

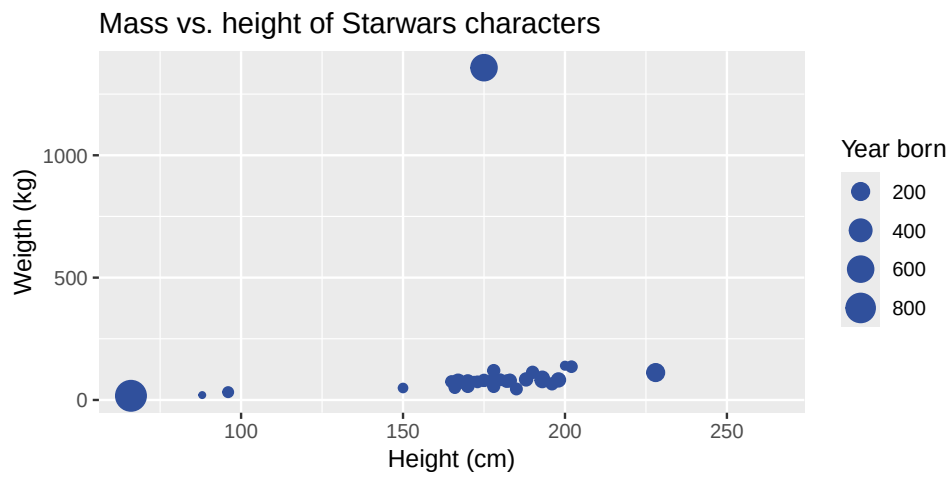
```
starwars
# # A tibble: 87 x 14
#   name      height  mass hair_color skin_color eye_color birth_year sex  gender
#   <chr>      <int> <dbl> <chr>      <chr>      <chr>      <dbl> <chr> <chr>
```

```
# 1 Luke Sk~ 172 77 blond fair blue 19 male mascu~
# 2 C-3P0 167 75 <NA> gold yellow 112 none mascu~
# 3 R2-D2 96 32 <NA> white, bl~ red 33 none mascu~
# 4 Darth V~ 202 136 none white yellow 41.9 male mascu~
# 5 Leia Or~ 150 49 brown light brown 19 fema~ femin~
# 6 Owen La~ 178 120 brown, gr~ light blue 52 male mascu~
# 7 Beru Wh~ 165 75 brown light blue 47 fema~ femin~
# 8 R5-D4 97 32 <NA> white, red red NA none mascu~
# 9 Biggs D~ 183 84 black light brown 24 male mascu~
# 10 Obi-Wan~ 182 77 auburn, w~ fair blue-gray 57 male mascu~
# # i 77 more rows
# # i 5 more variables: homeworld <chr>, species <chr>, films <list>,
# # vehicles <list>, starships <list>
```

```
a) ggplot(starwars,
  aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "pink")
```

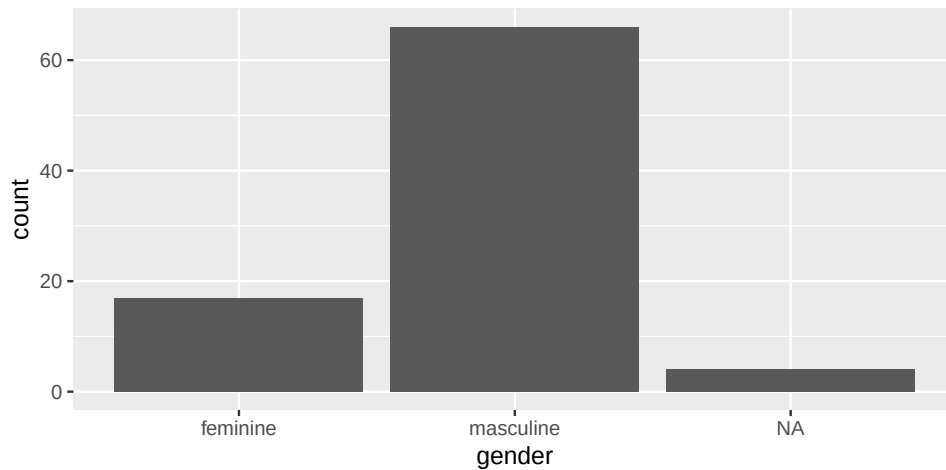


```
b) ggplot(starwars,
  aes(x = height, y = mass, size = birth_year)) +
  geom_point(color = "#30509C") +
  labs(
    title = "Mass vs. height of Starwars characters",
    x = "Height (cm)",
    y = "Weigth (kg)",
    size = "Year born"
  )
```



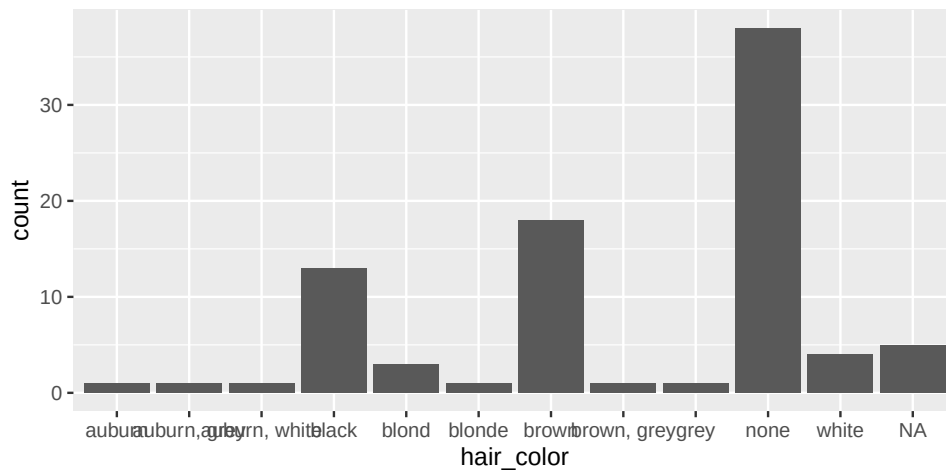
c) We can have a look at the distribution of **gender**

```
ggplot(starwars, aes(x = gender)) +  
  geom_bar()
```



or **hair_color**

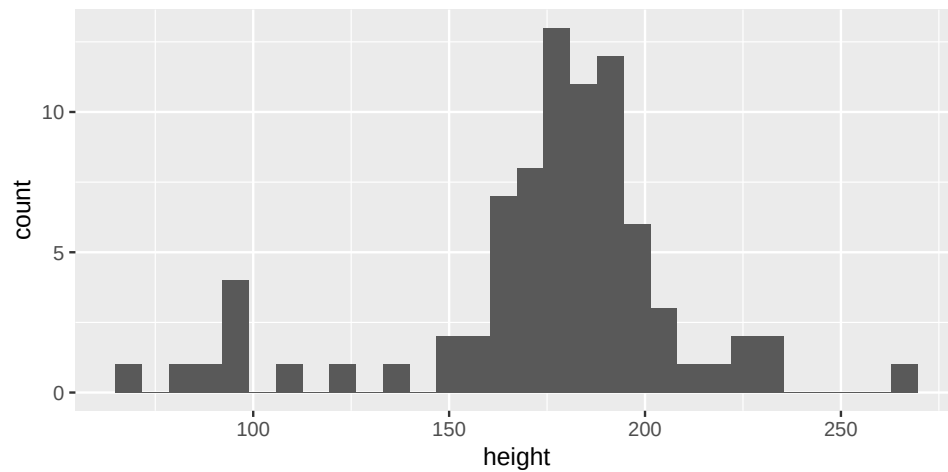
```
ggplot(starwars, aes(x = hair_color)) +  
  geom_bar()
```



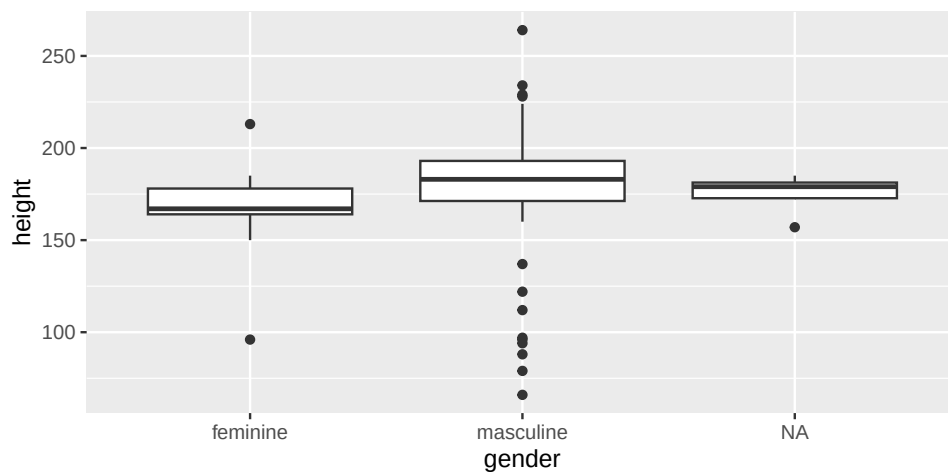
The other categorical variables have quite a number of levels.

- d) The dataset contains the numeric variables `height`, `mass`, and `birth_year`. We pick `height` for your histogram.

```
ggplot(starwars, aes(x = height)) +  
  geom_histogram()
```

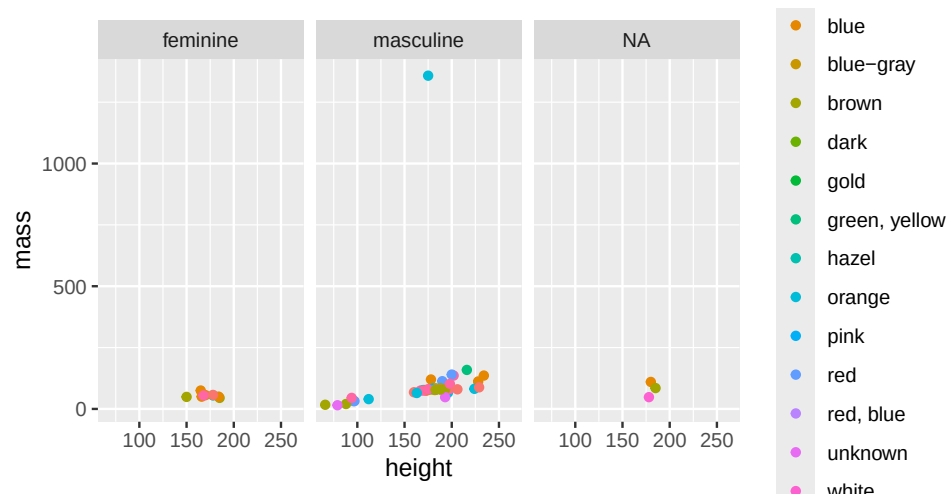


- e) `ggplot(starwars, aes(x = gender, y = height)) +
 geom_boxplot()`



Masculine characters have, on average, a larger height than feminine characters. The variation of height is also much larger in the class of masculine characters than in the class of feminine characters.

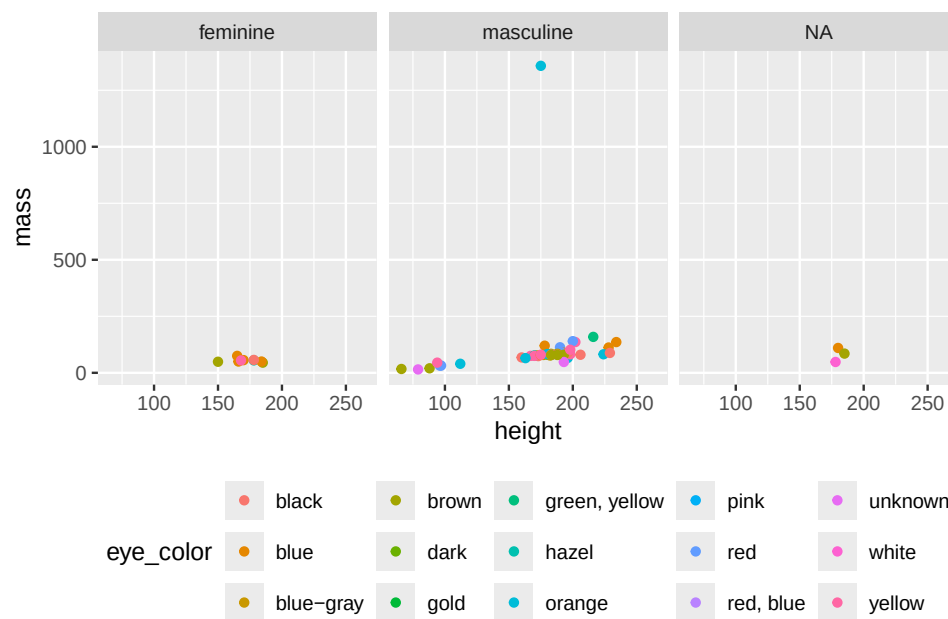
- f) `ggplot(starwars, aes(x = height, y = mass, colour = eye_color)) +
 geom_point() +
 facet_wrap(~ gender)`



None of the two variables, `mass` and `height`, seem to depend on the eye color. Moreover, the relation between `mass` and `height` seems not to be affected by eye color.

Remark: The legend is rather hard to read due to the large number of levels for `eye_color`. To improve this situation, we can change the position of the legend to "bottom".

```
ggplot(starwars, aes(x = height, y = mass, colour = eye_color)) +
  geom_point() +
  facet_wrap(~ gender) +
  theme(legend.position = "bottom")
```



Remark: Use the Quarto file `hotels-template.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 2.2.

```
library(tidyverse)
```

The dataset used in this exercise comes from Tidy Tuesday. It's data about hotel bookings. Since the data is not contained in some package, we have to read in the data on our own. To do so, we use the `read_csv()` function from the `readr` (part of the `tidyverse`) package.

```
hotels <- read_csv(
  "https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-02-11/hotels.csv")
```

Remark. The `read_csv()` function works not only if the data file is stored on your computer, you can also use it to read csv files that are available on a web server.

- a) Warm up! Take a look at an overview of the data with the `summary()` function.

Note

I already gave you the answer to this exercise. You only need to render the document and view the output. All variables are defined in the Data Dictionary, which is part of the template `hotels-template.qmd`. Thus, you don't need to familiarize yourself with all variables to work through these exercises.

```
summary(hotels)
```

- b) Are people traveling on a whim? Let's see...

Fill in the blanks for filtering for hotel bookings where the guest is **not** from the US (country code "USA") and the `lead_time` is less than one day.

Note

You will need to set `eval: true` when you have an answer you want to try out.

```
hotels |>
  filter(
    country ____ "USA",
    lead_time ____ ____
  )
```

- c) How many bookings involve at least one child **or** baby?

In the following chunk, replace

- [AT LEAST] with the logical operator for "at least" (in two places)
- [OR] with the logical operator for "or"

```
hotels |>
  filter(
    children [AT LEAST] 1 [OR] babies [AT LEAST] 1
  )
```

- d) Do you think it's more likely to find bookings with children or babies in city hotels or resort hotels? Test your intuition. Use `filter()` to determine the number of bookings in resort hotels with at least one child **or** baby in the room. Then, do the same for city hotels and compare the numbers of rows in the resulting filtered data frames.

```
# add code here
# pay attention to correctness and code style
```

```
# add code here
# pay attention to correctness and code style
```

- e) Create a frequency table of the number of `adults` in a booking. Display the results in descending order so the most common observation is on top. What is the most common number of adults in bookings in this dataset? Are there any surprising results?

Note

Don't forget to label your R chunk as well (where it says `label-me-1`). Your label should be short, informative, and shouldn't include spaces. It also shouldn't repeat a previous label, otherwise Quarto will give you an error about repeated R chunk labels.

```
# add code here
# pay attention to correctness and code style
```

- f) Repeat e) once for canceled bookings (`is_canceled` coded as 1) and once for not canceled bookings (`is_canceled` coded as 0). What does this reveal about the surprising results you spotted in the previous exercise?

```
# add code here
# pay attention to correctness and code style
```

Solution:

a) `summary(hotels)`

```
#      hotel      is_canceled      lead_time      arrival_date_year
# Length:119390   Min.      :0.0000   Min.      : 0   Min.      :2015
# Class :character 1st Qu.:0.0000   1st Qu.: 18   1st Qu.:2016
# Mode  :character Median :0.0000   Median : 69   Median :2016
#              Mean  :0.3704   Mean  :104   Mean  :2016
#              3rd Qu.:1.0000   3rd Qu.:160   3rd Qu.:2017
#              Max.  :1.0000   Max.  :737   Max.  :2017
#
# arrival_date_month arrival_date_week_number arrival_date_day_of_month
# Length:119390      Min.      : 1.00      Min.      : 1.0
# Class :character   1st Qu.:16.00      1st Qu.: 8.0
# Mode  :character   Median :28.00      Median :16.0
#              Mean  :27.17      Mean  :15.8
#              3rd Qu.:38.00      3rd Qu.:23.0
#              Max.  :53.00      Max.  :31.0
#
# stays_in_weekend_nights stays_in_week_nights      adults
# Min.      : 0.0000      Min.      : 0.0      Min.      : 0.000
```

```

# 1st Qu.: 0.0000      1st Qu.: 1.0      1st Qu.: 2.000
# Median : 1.0000      Median : 2.0      Median : 2.000
# Mean   : 0.9276      Mean   : 2.5      Mean   : 1.856
# 3rd Qu.: 2.0000      3rd Qu.: 3.0      3rd Qu.: 2.000
# Max.    :19.0000      Max.    :50.0      Max.    :55.000
#
#      children      babies      meal      country
# Min.   : 0.0000    Min.   : 0.000000    Length:119390    Length:119390
# 1st Qu.: 0.0000    1st Qu.: 0.000000    Class :character    Class :character
# Median : 0.0000    Median : 0.000000    Mode  :character    Mode  :character
# Mean   : 0.1039    Mean   : 0.007949
# 3rd Qu.: 0.0000    3rd Qu.: 0.000000
# Max.   :10.0000    Max.   :10.000000
# NA's    :4
# market_segment    distribution_channel    is_repeated_guest
# Length:119390      Length:119390      Min.   :0.00000
# Class :character    Class :character    1st Qu.:0.00000
# Mode  :character    Mode  :character    Median :0.00000
#                                     Mean   :0.03191
#                                     3rd Qu.:0.00000
#                                     Max.   :1.00000
#
# previous_cancellations    previous_bookings_not_canceled    reserved_room_type
# Min.   : 0.00000          Min.   : 0.0000          Length:119390
# 1st Qu.: 0.00000          1st Qu.: 0.0000          Class :character
# Median : 0.00000          Median : 0.0000          Mode  :character
# Mean   : 0.08712          Mean   : 0.1371
# 3rd Qu.: 0.00000          3rd Qu.: 0.0000
# Max.   :26.00000          Max.   :72.0000
#
# assigned_room_type    booking_changes    deposit_type      agent
# Length:119390          Min.   : 0.0000    Length:119390      Length:119390
# Class :character        1st Qu.: 0.0000    Class :character    Class :character
# Mode  :character        Median : 0.0000    Mode  :character    Mode  :character
#                                     Mean   : 0.2211
#                                     3rd Qu.: 0.0000
#                                     Max.   :21.0000
#
#      company      days_in_waiting_list    customer_type      adr
# Length:119390      Min.   : 0.000    Length:119390      Min.   : -6.38
# Class :character    1st Qu.: 0.000    Class :character    1st Qu.: 69.29
# Mode  :character    Median : 0.000    Mode  :character    Median : 94.58
#                                     Mean   : 2.321          Mean   : 101.83
#                                     3rd Qu.: 0.000          3rd Qu.: 126.00
#                                     Max.   :391.000        Max.   :5400.00
#
# required_car_parking_spaces    total_of_special_requests    reservation_status
# Min.   :0.00000          Min.   :0.0000          Length:119390
# 1st Qu.:0.00000          1st Qu.:0.0000          Class :character
# Median :0.00000          Median :0.0000          Mode  :character
# Mean   :0.06252          Mean   :0.5714
# 3rd Qu.:0.00000          3rd Qu.:1.0000
# Max.   :8.00000          Max.   :5.0000

```



```
#
# reservation_status_date
# Min. :2014-10-17
# 1st Qu.:2016-02-01
# Median :2016-08-07
# Mean :2016-07-30
# 3rd Qu.:2017-02-08
# Max. :2017-09-14
#
```

b) `hotels |>`
`filter(`
`country != "USA",`
`lead_time < 1`
`)`

```
# # A tibble: 6,174 x 32
#   hotel          is_canceled lead_time arrival_date_year arrival_date_month
#   <chr>          <dbl>      <dbl>          <dbl> <chr>
# 1 Resort Hotel      0          0            2015 July
# 2 Resort Hotel      0          0            2015 July
# 3 Resort Hotel      0          0            2015 July
# 4 Resort Hotel      0          0            2015 July
# 5 Resort Hotel      0          0            2015 July
# 6 Resort Hotel      0          0            2015 July
# 7 Resort Hotel      0          0            2015 July
# 8 Resort Hotel      0          0            2015 July
# 9 Resort Hotel      0          0            2015 July
#10 Resort Hotel      0          0            2015 July
# # i 6,164 more rows
# # i 27 more variables: arrival_date_week_number <dbl>,
# #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
# #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
# #   meal <chr>, country <chr>, market_segment <chr>,
# #   distribution_channel <chr>, is_repeated_guest <dbl>,
# #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>, ...
```

c) `hotels |>`
`filter(`
`children >= 1 | babies >= 1`
`)`

```
# # A tibble: 9,332 x 32
#   hotel          is_canceled lead_time arrival_date_year arrival_date_month
#   <chr>          <dbl>      <dbl>          <dbl> <chr>
# 1 Resort Hotel      0         18            2015 July
# 2 Resort Hotel      1         47            2015 July
# 3 Resort Hotel      0          1            2015 July
# 4 Resort Hotel      0         10            2015 July
# 5 Resort Hotel      1         79            2015 July
# 6 Resort Hotel      0        101            2015 July
# 7 Resort Hotel      0         92            2015 July
# 8 Resort Hotel      1         26            2015 July
# 9 Resort Hotel      0        102            2015 July
```

```
# 10 Resort Hotel          0          78          2015 July
# # i 9,322 more rows
# # i 27 more variables: arrival_date_week_number <dbl>,
# #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
# #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
# #   meal <chr>, country <chr>, market_segment <chr>,
# #   distribution_channel <chr>, is_repeated_guest <dbl>,
# #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>, ...
```

d) `hotels |>`

```
  filter(
    children >= 1 | babies >= 1, hotel == "Resort Hotel"
  )
# # A tibble: 3,929 x 32
#   hotel          is_canceled lead_time arrival_date_year arrival_date_month
#   <chr>          <dbl>      <dbl>          <dbl> <chr>
# 1 Resort Hotel      0         18          2015 July
# 2 Resort Hotel      1         47          2015 July
# 3 Resort Hotel      0          1          2015 July
# 4 Resort Hotel      0         10          2015 July
# 5 Resort Hotel      1         79          2015 July
# 6 Resort Hotel      0        101          2015 July
# 7 Resort Hotel      0         92          2015 July
# 8 Resort Hotel      1         26          2015 July
# 9 Resort Hotel      0        102          2015 July
#10 Resort Hotel      0         78          2015 July
# # i 3,919 more rows
# # i 27 more variables: arrival_date_week_number <dbl>,
# #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
# #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
# #   meal <chr>, country <chr>, market_segment <chr>,
# #   distribution_channel <chr>, is_repeated_guest <dbl>,
# #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>, ...
```

```
hotels |>
  filter(
    children >= 1 | babies >= 1, hotel == "City Hotel"
  )
# # A tibble: 5,403 x 32
#   hotel          is_canceled lead_time arrival_date_year arrival_date_month
#   <chr>          <dbl>      <dbl>          <dbl> <chr>
# 1 City Hotel        1        100          2015 July
# 2 City Hotel        0         69          2015 July
# 3 City Hotel        0         67          2015 July
# 4 City Hotel        1         60          2015 July
# 5 City Hotel        0          0          2015 July
# 6 City Hotel        0          0          2015 July
# 7 City Hotel        0          0          2015 July
# 8 City Hotel        0         84          2015 July
# 9 City Hotel        0         74          2015 July
#10 City Hotel        0         75          2015 July
# # i 5,393 more rows
```

```
# # i 27 more variables: arrival_date_week_number <dbl>,
# #   arrival_date_day_of_month <dbl>, stays_in_weekend_nights <dbl>,
# #   stays_in_week_nights <dbl>, adults <dbl>, children <dbl>, babies <dbl>,
# #   meal <chr>, country <chr>, market_segment <chr>,
# #   distribution_channel <chr>, is_repeated_guest <dbl>,
# #   previous_cancellations <dbl>, previous_bookings_not_canceled <dbl>, ...
```

We have more observations in city hotels than in resort hotels. But overall we have

```
table(hotels$hotel)
#
#   City Hotel Resort Hotel
#       79330       40060
```

also almost twice as many bookings from city hotels.

e) `hotels |>`
`count(adults, sort = TRUE)`

```
# # A tibble: 14 x 2
#   adults      n
#   <dbl> <int>
# 1      2 89680
# 2      1 23027
# 3      3  6202
# 4      0   403
# 5      4    62
# 6     26     5
# 7      5     2
# 8     20     2
# 9     27     2
# 10     6     1
# 11    10     1
# 12    40     1
# 13    50     1
# 14    55     1
```

f) `hotels |>`
`filter(is_canceled == 1) |>`
`count(adults, sort = TRUE)`

```
# # A tibble: 14 x 2
#   adults      n
#   <dbl> <int>
# 1      2 35258
# 2      1  6674
# 3      3  2151
# 4      0   109
# 5      4    16
# 6     26     5
# 7      5     2
# 8     20     2
# 9     27     2
# 10     6     1
```

```
# 11      10      1
# 12      40      1
# 13      50      1
# 14      55      1

hotels |>
  filter(is_canceled == 0) |>
  count(adults, sort = TRUE)
# # A tibble: 5 x 2
#   adults      n
#   <dbl> <int>
# 1      2 54422
# 2      1 16353
# 3      3  4051
# 4      0   294
# 5      4    46
```

3 Explore data

Exercise 3.1. Consider the American Community Survey (2012). Data from this survey is contained in the data set `acs12` from the `openintro` package.

- Use R to compute the mean and median of the variable `time to work` for each level of the variable `race`.
- What kind of shape has the distribution of `time_to_work` for each level of `race`? Answer this question by producing boxplots of `time_to_work` for each level of `race`.
- Analyse the association between travel time to work and income. Explain your findings.

Solution:

- Computing sample statistics for each dataset group separately can be done with functions from the `dplyr` package - as we have seen in the Data Wrangling part. In this case, the sample statistics mean, and median should be computed for each level of the variable `race`. Hence, we first have to group the data `acs12` with respect to `race`. Afterwards, we can summarize the data by computing the mean and median.

In the first step, we load the data set, contained in the `openintro` package, and the `tidyverse`.

```
library(openintro)
library(tidyverse)
```

Now we could try the following

```
acs12 |>
  group_by(race) |>
  summarise(mean_tw = mean(time_to_work),
            median_tw = median(time_to_work))
# # A tibble: 4 x 3
#   race mean_tw median_tw
#   <fct>   <dbl>    <int>
# 1 white      NA        NA
# 2 black      NA        NA
# 3 asian      NA        NA
# 4 other      NA        NA
```

which is not the desired result. All values are NA (not available - missing). This is because for each level of `race`, there is at least one missing value of `time_to_work`. Whenever at least one input value is missing, the default value of the sample statistic will be NA. But this default behavior can be changed for `mean()` and `median()` by setting the argument `na.rm` (remove the missing values prior to compute the sample statistic) equal to `TRUE`. This solution also works for several other functions. Using `na.rm = TRUE`, then leads to

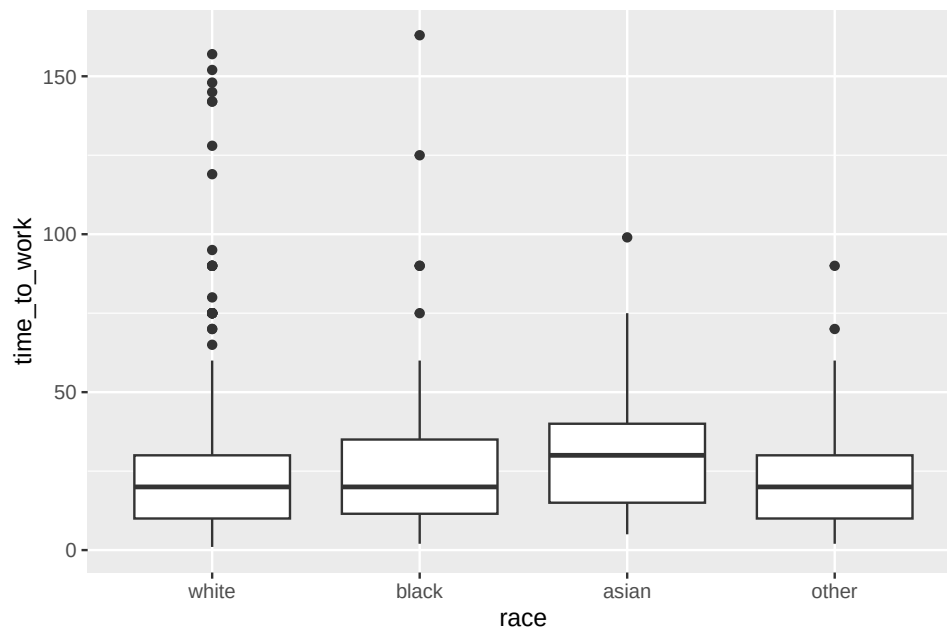
```
acs12 |>
  group_by(race) |>
  summarise(mtw = mean(time_to_work, na.rm = TRUE),
            metw = median(time_to_work, na.rm = TRUE))
# # A tibble: 4 x 3
#   race   mtw metw
#   <fct> <dbl> <dbl>
# 1 white  25.4   20
```

```
# 2 black 29.7 20
# 3 asian 31.5 30
# 4 other 23.8 20
```

We detect that the mean is higher than the median for all four levels.

- b) The values in a) suggest that distributions for **white**, **black** and **other** might be right-skewed. The distribution for **asian** on the other hand might be symmetric. For **white** and **black** the assumption is confirmed by the box plots.

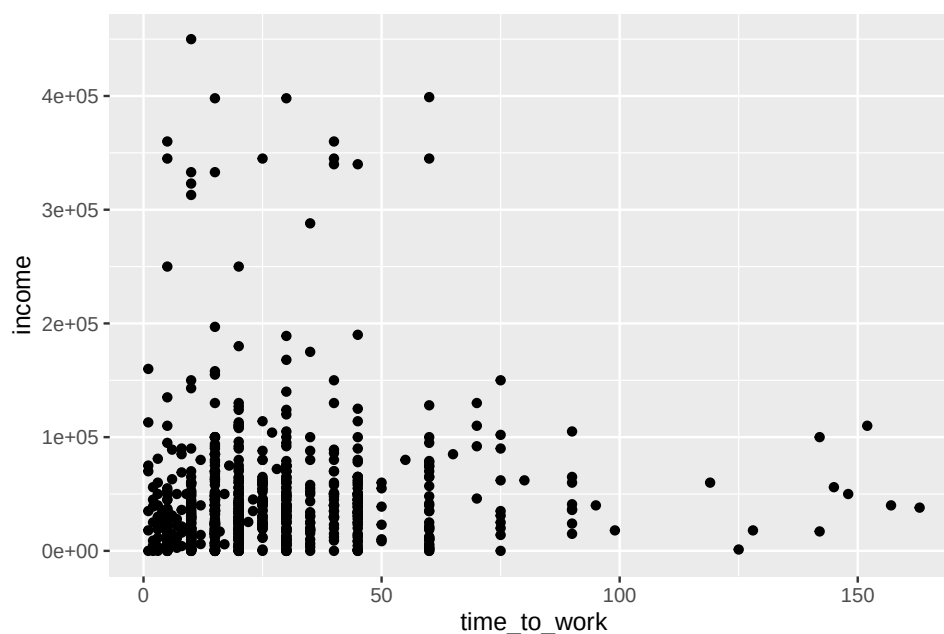
```
ggplot(acs12, aes(x=race, y=time_to_work)) +
  geom_boxplot()
```



For **asian**, the distribution looks slightly left-skewed (when focusing on the box). The mean being larger than the median is just due to the one outlier. The distribution for **other** seems slightly right-skewed.

- c) The variables **time_to_work** and **income** are numeric. We therefore create a scatterplot to visualize the association. This can be done by using **ggplot()** in combination with **geom_point()**.

```
ggplot(acs12, aes(x = time_to_work, y = income)) +
  geom_point()
```



From the plot, we can see that the larger annual incomes - defined, e.g., by being larger than the 95-th percentile are observed just for the smaller (less than 50 minutes) travel times to work.

```
quantile(acs12$income, .95, na.rm = TRUE)
# 95%
# 92000
```

On the other hand, annual incomes less than the 95th percentile are observed over the whole range of observed travel times to work. However, within these two subgroups, no clear association between travel time to work and annual income can be detected.

Exercise 3.2. A company surveyed 910 randomly sampled customers at a large shopping mall about their preferred type of reusable water bottle. Respondents could choose:

- (i) Stainless steel
- (ii) Plastic (BPA-free)
- (iii) Glass
- (iv) Not sure

The results are shown by age group:

	<i>Age group</i>			Total
	18–29	30–49	50+	
<i>(i)</i> Stainless steel	84	112	82	278
<i>(ii)</i> Plastic (BPA-free)	103	118	41	262
<i>(iii)</i> Glass	132	98	120	350
<i>(iv)</i> Not sure	7	5	8	20
Total	326	333	251	910

- a) What percent of these shoppers are in the 18–29 age group?
- b) What percent of these shoppers prefer stainless steel bottles?
- c) What percent of these shoppers are 18–29 and prefer stainless steel?
- d) Among each age group, what percent prefers stainless steel?
- e) Do age group and bottle preference appear to be independent? Explain.

Solution:

- a) The proportion of shoppers age 18–29 is

$$\frac{326}{910} \approx 0.358 \text{ or } 35.8\%.$$

- b) The proportion of shoppers who prefer stainless steel bottles is

$$\frac{278}{910} \approx 0.305 \text{ or } 30.5\%.$$

- c) The proportion who are 18–29 and prefer stainless steel is

$$\frac{84}{910} \approx 0.092 \text{ or } 9.2\%.$$

- d) Percent in each age group who prefer stainless steel:

$$18\text{--}29: \frac{84}{326} \approx 0.258 \text{ or } 25.8\%,$$

$$30\text{--}49: \frac{112}{333} \approx 0.336 \text{ or } 33.6\%,$$

$$50+: \frac{82}{251} \approx 0.327 \text{ or } 32.7\%.$$

- e) Since the percentages differ across age groups (especially the lower stainless-steel preference among 18–29 shoppers), the variables *age group* and *bottle preference* do not appear to be independent.

Remark: Use the Quarto file `starwars-template-2.qmd`, which you find on Moodle in the folder "Data files, Quarto files and previous problem sets", to work on the following exercise.

Exercise 3.3. We want to work with the `starwars` dataset, which is part of the `dplyr` package. Since we also need `ggplot2`, we load the `tidyverse`.

```
library(tidyverse)
```

- a) Visualize the distribution of the `height`, `mass` and `birth_year`. Describe the shape for each of the three distributions. Complete the following code.

```
ggplot(starwars, aes(x = height)) +  
  ---  
  
ggplot(starwars, aes(x = mass)) +  
  ---  
  
ggplot(starwars, aes(x = birth_year)) +  
  ---
```

- b) Visualize the relation between the variables `mass` and `height`, including the information about `hair_color` and `gender`. Provide an interpretation with your answer. Use the function `drop_na()` to remove observations that miss `gender` and `hair_color` information when creating the plot.

```
# add code here  
# pay attention to correctness and code style
```

- c) Calculate minimum, mean, median, and maximum height grouped by the actual sex of the character. Are men larger on average?

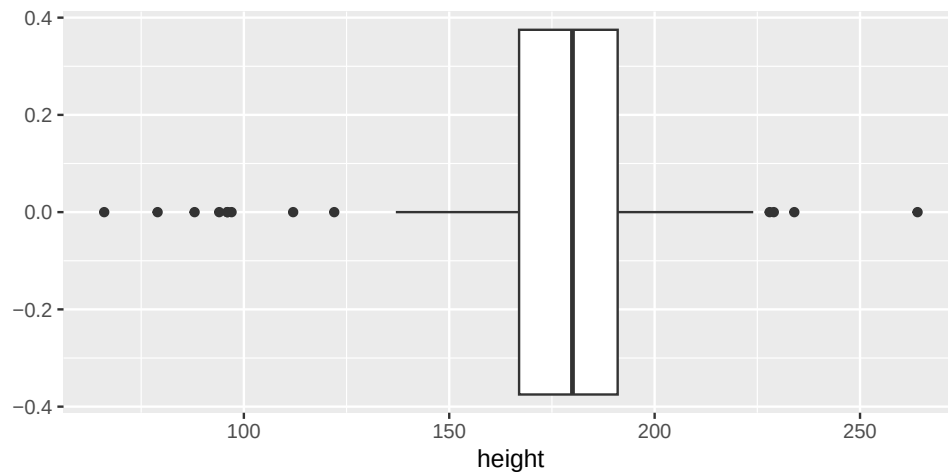
Hint: be aware that there are missing values in the data set.

```
# add code here  
# pay attention to correctness and code style
```

Solution:

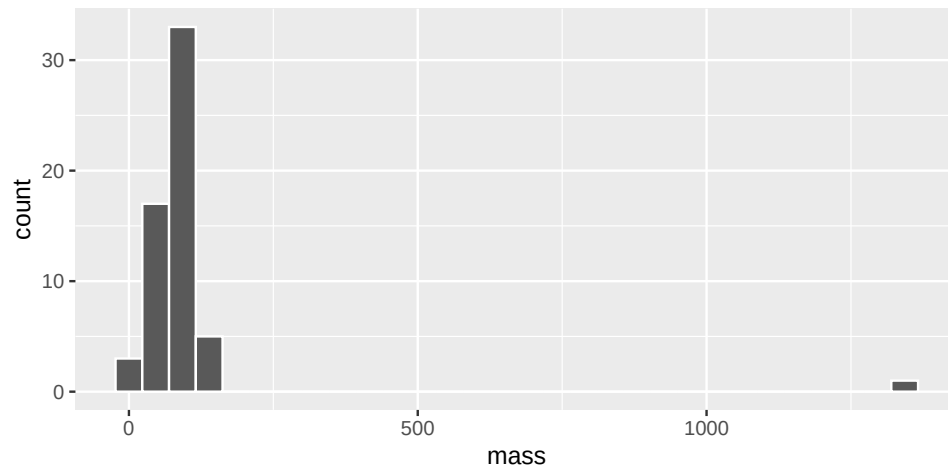
- a) All three variables are numeric. The sample size is not that small, so a dot plot is inappropriate. But we could create a histogram or boxplot. For each variable, we do either one of them.

```
ggplot(starwars, aes(x = height)) +  
  geom_boxplot()
```



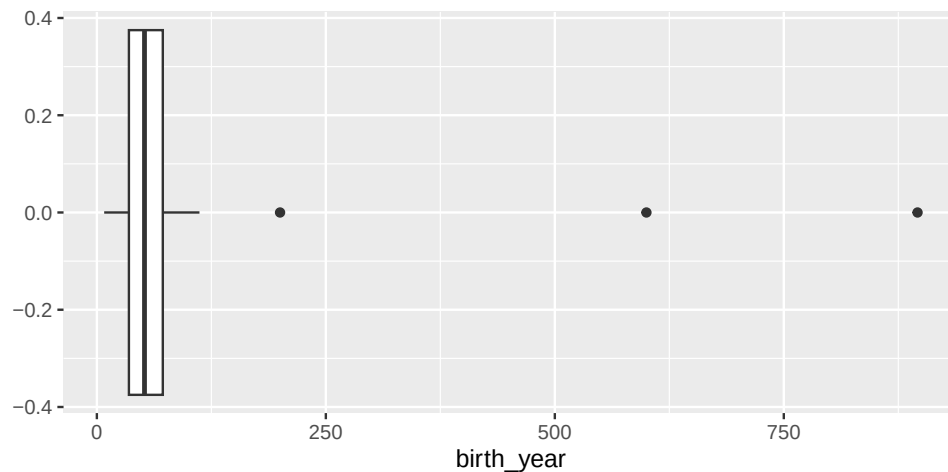
The distribution of height is relatively symmetric. The location of the median is almost the center of the box. But we detect several rather large as well as small observations.

```
ggplot(starwars, aes(x = mass)) +  
  geom_histogram(colour = "white")
```



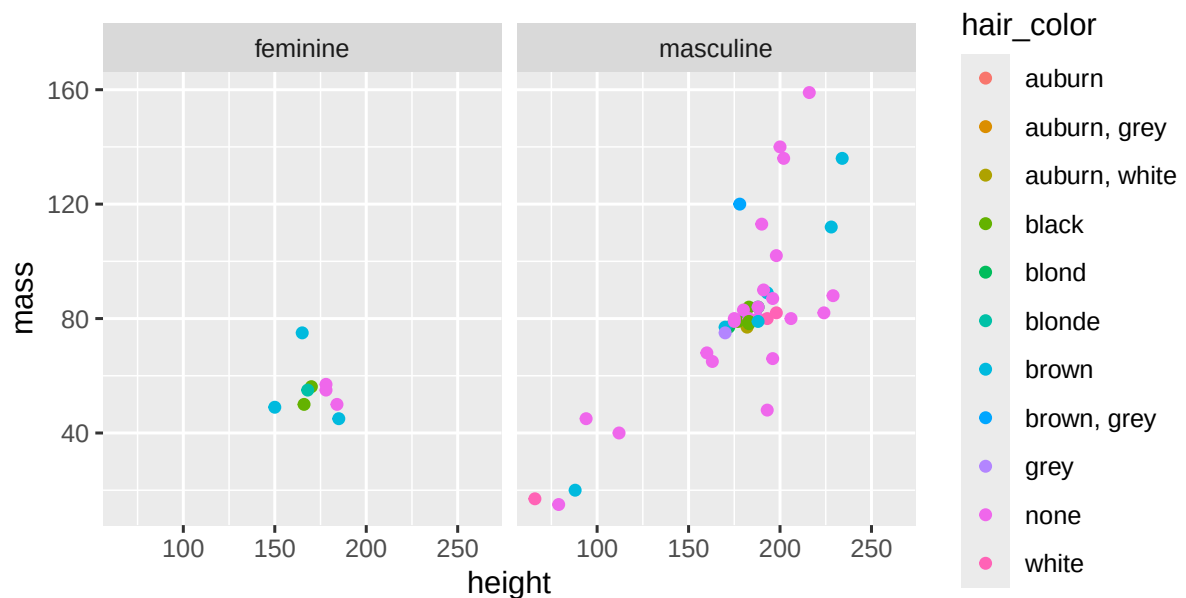
The shape of the distribution is influenced by some large (one very large) observations. This leads to a right-skewed distribution. But it's a bit hard to say, just based on the histogram.

```
ggplot(starwars, aes(x = birth_year)) +  
  geom_boxplot()
```



The shape of the birth year distribution is again relatively symmetric, although there are three outliers present.

```
b) ggplot(drop_na(starwars, gender, hair_color),
  aes(x = height, y = mass, colour = hair_color)) +
  geom_point() +
  facet_wrap(~ gender)
```



The height of feminine characters is very much concentrated in the sample. Therefore, it's hard to say whether the two variables are associated. For the masculine characters, we can detect a relatively strong positive dependence. But since there are no clear clusters due to color, we can't say that mass and height vary with respect to hair color (which is also not surprising).

```
c) starwars |>
  group_by(sex) |>
  summarize(min_h = min(height, na.rm = TRUE),
    mean_h = mean(height, na.rm = TRUE),
    med_h = median(height, na.rm = TRUE),
    max_h = max(height, na.rm = TRUE),
    n())
```

A tibble: 5 x 6

#	sex	min_h	mean_h	med_h	max_h	`n()`
#	<chr>	<int>	<dbl>	<dbl>	<int>	<int>
# 1	female	150	172.	168.	213	16
# 2	hermaphroditic	175	175	175	175	1
# 3	male	66	179.	183	264	60
# 4	none	96	131.	97	200	6
# 5	<NA>	157	175	179	185	4

We argued before that the distribution of height is relatively symmetric. Therefore, the median and empirical mean both describe the center of the distribution (maybe not valid for the Droids - **none**). Considering just observations with known sex, males have the largest height values on average. Just the four observations with unknown **sex** seem to be larger on average.

Exercise 3.4. Consider the following paired numerical data describing the weekly study hours (**hours**) and exam scores (**score**) of 12 students:

```
df <- tibble(  
  hours = c(6, 8, 3, 11, 7, 12, 9, 4, 10, 5, 6, 9),  
  score = c(59, 66, 45, 78, 63, 80, 75, 49, 72, 55, 63, 70),  
  student = 1:12  
)
```

- a) Create a scatterplot of **hours** (x-axis) and **score** (y-axis). Describe whether a *linear* relationship appears plausible.
- i) Compute the empirical correlation coefficient $r_{(x,y)} = \frac{s_{xy}}{s_x s_y}$ and interpret the result in terms of direction, strength, and linearity.

Note: Don't use R (the function `cor()`) to compute the correlation. Use instead the below given values and the formula defining the correlation coefficient.

```
sum((df$hours - mean(df$hours)) * (df$score - mean(df$score)))  
# [1] 337.5  
sd(df$hours)  
# [1] 2.812311  
sd(df$score)  
# [1] 11.14743
```

- b) Repeat the calculation after replacing the score of the 6th student with 55 (keeping **hours** = 12). How does $r_{(x,y)}$ change? What does this illustrate about the sensitivity of the Pearson correlation?

Hint: Create a new dataset `df_extreme` where you change the score of the 6th student. Then compute the correlation using `cor()`.

- c) Suppose the students belong to two study programs. Students 1 to 6 belong to program A and the rest to program B. Add a variable **program** to `df`, and then compute the correlation separately for programs A and B. Compare the groupwise correlations to the overall correlation and comment.

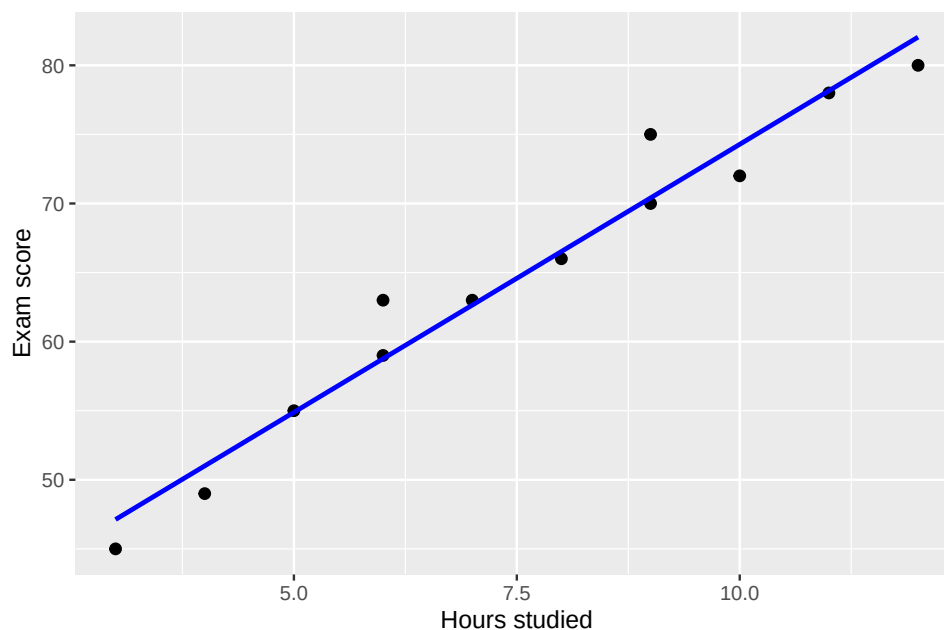
Solution:

We start by copying the data from the problem into our R session.

```
library(tidyverse)  
df <- tibble(  
  hours = c(6, 8, 3, 11, 7, 12, 9, 4, 10, 5, 6, 9),  
  score = c(59, 66, 45, 78, 63, 80, 75, 49, 72, 55, 63, 70)  
)
```

- a) A scatterplot of study hours vs. exam score can be created using the following code.

```
ggplot(df, aes(x = hours, y = score)) +  
  geom_point(size = 2) +  
  geom_smooth(method = "lm", se = FALSE, color = "blue") +  
  labs(x = "Hours studied", y = "Exam score")
```



The cloud of points suggests a positive linear trend: as hours increase, the score tends to increase roughly linearly.

b) Now we want to compute the correlation. From the lecture we know that

$$r_{(x,y),n} = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{\sqrt{\sum_{i=1}^n (x_i - \bar{x}_n)^2 \sum_{i=1}^n (y_i - \bar{y}_n)^2}} = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{(n-1)s_{x,n}s_{y,n}}$$

$$= \frac{s_{xy,n}}{s_{x,n}s_{y,n}}.$$

Using the values given in the note, we can compute

$$r_{(x,y),n} = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)(y_i - \bar{y}_n)}{(n-1)s_{x,n}s_{y,n}} = \frac{337.5}{(12-1) \cdot 2.812311 \cdot 11.14743}$$

$$\approx 0.9787,$$

Only to check if we computed the correct value:

```
cor(df$hours, df$score)
# [1] 0.9786854
```

Interpretation:

- Direction: positive
- Strength: very strong (close to 1)
- Linearity: consistent with a clear linear association

Substantive conclusion: students who study more tend to obtain substantially higher exam scores.

c) We first create a new tibble by assigning `df` to a new object.

```
df_extreme <- df
```

Now we replace the score of the 6th student by 55 and compute the correlation.

```
df_extreme$score[6] <- 55
cor(df_extreme$hours, df_extreme$score)
# [1] 0.7055302
```

The correlation becomes much smaller, when decreasing the score of the 6th student. The Pearson correlation is sensitive to outliers because it measures linear covariation using means and Euclidean distances. A single extreme point can substantially change the value.

- d) The first step is to create the **program** variable. This can be done using `mutate()`:

```
df <- df |>
  mutate(program = rep(c("A","B"), each = 6))
```

or using the ' \$ ' notation again:

```
df$program <- rep(c("A","B"), each = 6)
```

For the group correlations, we obtain the following values.

```
df |>
  group_by(program) |>
  summarize(r = cor(hours, score))
# # A tibble: 2 x 2
#   program      r
#   <chr>    <dbl>
# 1 A      0.996
# 2 B      0.956
```

The linear association in program A is even stronger compared to the whole sample. Hence, it must have the opposite effect for program B.

4 Probability

Exercise 4.1. Assortative mating is a non-random mating pattern where individuals with similar genotypes and/or phenotypes mate more frequently than expected under a random mating pattern. Researchers studying this topic collected data on the eye colors of 204 Scandinavian men and their female partners. The table below summarizes the results.

		<i>Partner (female)</i>			Total
		Blue	Brown	Green	
<i>Self (male)</i>	Blue	78	23	13	114
	Brown	19	23	12	54
	Green	11	9	16	36
	Total	108	55	41	204

- What is the probability that a randomly chosen male respondent or his partner has blue eyes?
- What is the probability that a randomly chosen male respondent with blue eyes has a partner with blue eyes?
- What is the probability that a randomly chosen male respondent with brown eyes has a partner with blue eyes? What is the probability of a randomly chosen male respondent with green eyes having a partner with blue eyes?
- Does the eye color of male respondents and their partners appear to be independent? Explain your reasoning.

Solution:

- $P(\text{man or partner has blue eyes}) = \frac{108+114-78}{204} \approx 0.7059$
- $P(\text{partner with blue eyes}|\text{man with blue eyes}) = \frac{78}{114} = 0.6842$
-

$$P(\text{partner with blue eyes}|\text{man with brown eyes}) = \frac{19}{54} \approx 0.3519$$

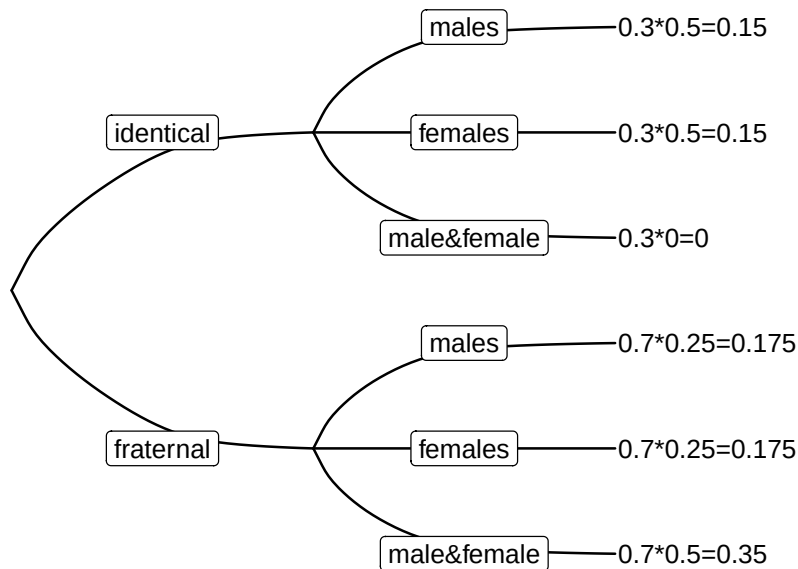
$$P(\text{partner with blue eyes}|\text{man with green eyes}) = \frac{11}{36} \approx 0.3056$$

- It is much more likely for a man with blue eyes to have a partner with blue eyes than a man with another eye color to have a partner with blue eyes. Therefore, the eye colors of males and their partners do not appear to be independent.

Exercise 4.2. About 30% of human twins are identical, and the rest are fraternal. Identical twins are necessarily the same sex – half are males, and the other half are females. One-quarter of fraternal twins are both males, one-quarter are both females, and one-half are mixed: one male, one female. You have just become a parent of twins and are told they are both girls. Given this information, what is the probability that they are identical?

Solution:

Let's draw a tree diagram.



We want to find the conditional probability $P(\text{identical}|\text{females})$, which is defined as we

$$P(\text{identical}|\text{females}) = \frac{P(\text{identical and females})}{P(\text{females})}.$$

From the tree diagram we get $P(\text{identical and females}) = 0.15$ and $P(\text{females}) = 0.15 + 0.175$, which then leads to

$$P(\text{identical}|\text{females}) = \frac{0.15}{0.15 + 0.175} \approx 0.46$$

Exercise 4.3. A genetic test is used to determine if people have a predisposition for *thrombosis*, which is the formation of a blood clot inside a blood vessel that obstructs blood flow through the circulatory system. It is believed that 3% of people have this predisposition. The genetic test is 99% accurate if a person has the predisposition, meaning that the probability of a positive test result when a person actually has the predisposition is 0.99. The test is 98% accurate if a person does not have the predisposition. What is the probability that a randomly selected person who tests positive for the predisposition by the test actually has the predisposition?

Solution:

We start by defining appropriate events. Let pre be the event of having predisposition and pos the event of positive test result. Then, we can write

$$P(\text{pre}) = 0.03, \quad P(\text{pos}|\text{pre}) = 0.99 \text{ and } P(\text{pos}^c|\text{pre}^c) = 0.98.$$

The aim is now to compute the probability $P(\text{pre}|\text{pos})$. We will use the Bayes theorem to calculate this probability.

$$\begin{aligned} P(\text{pre}|\text{pos}) &= \frac{P(\text{pre} \cap \text{pos})}{P(\text{pos})} = \frac{P(\text{pos}|\text{pre}) \cdot P(\text{pre})}{P(\text{pos}|\text{pre}) \cdot P(\text{pre}) + P(\text{pos}|\text{pre}^c) \cdot P(\text{pre}^c)} \\ &= \frac{0.99 \cdot 0.03}{0.99 \cdot 0.03 + (1 - 0.98) \cdot (1 - 0.03)} = 0.604888 \end{aligned}$$

Exercise 4.4. Consider the following card game with a well-shuffled standard 52-card deck. If you draw a red card, you win nothing. If you get a spade, you win \$5. For any club, you win \$10 plus an extra \$20 for the ace of clubs.

- a) Create a probability model for the amount you win at this game. Also, find the expected winnings for a single game and the standard deviation of the winnings.
- b) What is the maximum amount you would be willing to pay to play this game? Explain your reasoning.

Solution:

- a) The sample space for this random process is given by $S = \{0, 5, 10, 30\}$. The random variable X describes the random outcome of this game.

The probability distribution for this random process (game) is given by

$$P(X = 0) = \frac{26}{52}, P(X = 5) = \frac{13}{52}, P(X = 10) = \frac{12}{52}, P(X = 30) = \frac{1}{52}.$$

Now we are able to compute the expected value $E[X]$ (the expected winnings). It is given by

$$\begin{aligned} E[X] &= \sum_{x \in \{0, 5, 10, 30\}} x \cdot P(X = x) \\ &= 0 \cdot \frac{26}{52} + 5 \cdot \frac{13}{52} + 10 \cdot \frac{12}{52} + 30 \cdot \frac{1}{52} \approx 4.13. \end{aligned}$$

To compute the standard deviation, we first need to compute the variance.

$$\begin{aligned} \text{Var}[X] &= \sum_{x \in \{0, 5, 10, 30\}} (x - E[X])^2 \cdot P(X = x) \\ &= (0 - 4.13)^2 \cdot \frac{26}{52} + (5 - 4.13)^2 \cdot \frac{13}{52} + (10 - 4.13)^2 \cdot \frac{12}{52} + (30 - 4.13)^2 \cdot \frac{1}{52} \approx 29.54. \end{aligned}$$

This leads to a standard deviation of

$$\text{SD}[X] = \sqrt{\text{Var}[X]} = \sqrt{29.54} \approx 5.44.$$

- b) If you are playing with the goal of making money, you should not play unless the cost of the game is **less** than $E[X] \approx 4.13$.

Exercise 4.5. An airline charges the following baggage fees: \$25 for the first bag and \$35 for the second. Suppose 54% of passengers have no checked baggage, 34% have one piece of checked baggage, and 12% have two pieces. We suppose a negligible portion of people check more than two bags.

- Build a probability model, compute the average revenue per passenger, and compute the corresponding standard deviation.
- About how much revenue should the airline expect for a flight of 120 passengers? With what standard deviation? Note any assumptions you make and if you think they are justified.

Solution:

- The first step is to select a suitable probability model. To do so, we need to define a sample space S and, in addition, a probability for each outcome from the sample space S .

Let X be the baggage fee a passenger has to pay. Since the number of baggage is random, the fee can also be considered as a random variable. The sample space for this random variable is $S = \{0, 25, 60\}$, representing the fee for no luggage, one bag, and two bags, respectively. The probabilities for the three outcomes are

$$P(X = 0) = 0.54, \quad P(X = 25) = 0.34 \quad \text{and} \quad P(X = 60) = 0.12.$$

This defines our probability model for the random fee of a passenger.

The average revenue (from the airline's point of view) / fee (from the passengers' point of view) is then the expected value of X

$$E[X] = \sum_{x \in S} x \cdot P(X = x) = 0 \cdot P(X = 0) + 25 \cdot P(X = 25) + 60 \cdot P(X = 60) = 25 \cdot 0.34 + 60 \cdot 0.12 = 15.7$$

The next step is to compute the standard deviation. Since we get the standard deviation by taking the square root of the variance, we first have to compute the variance

$$\begin{aligned} \text{Var}[X] &= \sum_{x \in S} (x - E[X])^2 \cdot P(X = x) \\ &= (0 - 15.7)^2 \cdot P(X = 0) + (25 - 15.7)^2 \cdot P(X = 25) + (60 - 15.7)^2 \cdot P(X = 60) \\ &= 15.7^2 \cdot 0.54 + 9.3^2 \cdot 0.34 + 44.3^2 \cdot 0.12 = 398.01, \end{aligned}$$

and then the standard deviation

$$\text{SD}[X] = \sqrt{\text{Var}[X]} = \sqrt{398.01} = 19.950188.$$

- We assume independence between individual passengers. This **is probably not precisely correct**, but it would provide a helpful first approximation for the true revenue.

Let X_i be the fee passenger i pays with $i = 1, \dots, 120$. The average revenue of a flight with 120 passengers is then

$$E\left[\sum_{i=1}^{120} X_i\right] = \sum_{i=1}^{120} E(X_i) = \sum_{i=1}^{120} 15.7 = 120 \cdot 15.7 = 1884, .$$

To compute the standard deviation, we must first calculate the variance. Since we have to assume (no information is given about possible dependence between the passengers) independence

between the individual passengers, the variance of the sum of fees is just the sum of the variances of the individual fees

$$\text{Var} \left[\sum_{i=1}^{120} X_i \right] = \sum_{i=1}^{120} \text{Var}[X_i] = \sum_{i=1}^{120} 398.01 = 120 \cdot 398.01 = 4.77612 \times 10^4, .$$

The standard deviation is then obtained by taking the square root

$$\text{SD} \left[\sum_{i=1}^{120} X_i \right] = \sqrt{4.77612 \times 10^4} = 218.5433595 .$$

Exercise 4.6. During triathlons, participants are usually grouped by age and gender. We want to examine the time it takes to finish the triathlon. A suitable probability model for triathlon times is the normal distribution. However, each age and gender group will have its own distinct normal distribution. We will focus on the following two groups:

- Men, Ages 30-34: $\mathcal{N}(4313, 583^2)$
- Women, Ages 25-29: $\mathcal{N}(5261, 807^2)$

Times are listed in seconds. Use this information to compute each of the following:

- a) The cutoff time for the fastest 5% of athletes in the men's group, i.e., those who took the shortest 5% of time to finish.
- b) The cutoff time for the slowest 10% of athletes in the women's group.

Hint: Use R to compute any probability or quantile in this exercise.

Solution:

- a) The fastest 5% are in the 5th percentile of the distribution. So, we need to compute the 0.05 quantile of the $\mathcal{N}(4313, 583^2)$ distribution

```
qnorm(0.05, mean = 4313, sd = 583)
# [1] 3354.05
```

The fastest 5% of males in this age group finished in less than 56 minutes.

- b) The slowest 10% are in the 90th percentile of the distribution. So, we need to compute the 0.9 quantile of the $\mathcal{N}(5261, 807^2)$ distribution.

```
qnorm(0.9, mean = 5261, sd = 807)
# [1] 6295.212
```

The slowest 10% of females in this age group took more than 104 minutes to finish.

Exercise 4.7. The Capital Asset Pricing Model (CAPM) is a financial model that assumes returns on a portfolio are normally distributed. Let's consider a portfolio with an average annual return of 14.7% (i.e., an average gain of 14.7%) and a standard deviation of 33%. A 0% return indicates that the portfolio's value remains unchanged, a negative return signifies a loss, and a positive return indicates a gain.

- a) What percent of years does this portfolio lose money, i.e. have a return less than 0%?
- b) What is the cutoff for the highest 15% of annual returns with this portfolio?

Hint: Use R to compute any probability or quantile in this exercise.

Solution:

- a) Let X denote returns on this portfolio, then $X \sim \mathcal{N}(14.7, 33^2)$. The probability of losing money is then $P(X < 0)$, which is equal to

```
pnorm(0, mean = 14.7, sd = 33)
# [1] 0.3279957
```

- b) We are looking for the 85th percentile or 0.85 quantile. Using `qnorm()` we get the following result

```
qnorm(0.85, mean = 14.7, sd = 33)
# [1] 48.9023
```

5 Predictive modelling

Exercise 5.1. For each part a) through d), indicate whether we would generally expect the performance of a non-parametric statistical learning method to be better or worse than a parametric method. Justify your answer.

- a) The sample size n is extremely large, and the number of predictors p is small.
- b) The number of predictors p is extremely large, and the number of observations n is small.
- c) The relationship between the predictors and response is highly non-linear.
- d) The variance of the error terms, i.e., $\sigma^2 = \text{Var}(\epsilon)$, is extremely high.

Solution:

- a) *Expectation:* Non-parametric methods are generally expected to perform better.

Reason: With a large sample size, non-parametric methods can capture more complex relationships without overfitting because they have enough data to learn the underlying patterns. Parametric methods may be less flexible and could underfit if the true relationship is complex.

- b) *Expectation:* Parametric methods are generally expected to outperform non-parametric methods in this context.

Reason: Non-parametric methods are susceptible to the curse of dimensionality, making them prone to overfitting when the number of predictors is large and the sample size is small. Parametric methods, due to their simpler assumptions and fewer parameters, are more likely to generalize effectively in such situations, assuming the model is suitable for the data.

- c) *Expectation:* Non-parametric methods are expected to perform better.

Reason: Non-parametric methods do not impose a predetermined functional form on the relationship between predictors and response, enabling them to model complex, non-linear associations more effectively than parametric methods, which typically rely on linear or specific non-linear assumptions.

- d) *Expectation:* Parametric methods are generally expected to perform better.

Reason: High variance in the error terms increases the risk of overfitting for non-parametric methods, as they are more sensitive to noise. Parametric methods, due to their reduced flexibility, are generally more robust to high variance and less likely to overfit noisy data.

Exercise 5.2. Explain whether each scenario is a classification or regression problem, and indicate whether we are most interested in predicting an outcome variable or understanding the relation between the input variables and the output variable. Finally, provide the number n of distinct data points and the number of variables p that are available for use in making predictions.

- a) We collect data on the top 500 firms in the US. We record each firm's profit, number of employees, industry, and CEO salary. We are interested in understanding which factors affect CEO salary.
- b) We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product, we have recorded whether it was a success or a failure, the price charged, the marketing budget, the competitor's price, and 10 other variables.
- c) We are interested in predicting the % change in the USD/Euro exchange rate in relation to weekly changes in global stock markets. Hence, we collected weekly data for the entire year of 2012. For each week, we recorded the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Solution:

- a) This is a regression problem, as we want to understand the relations between the input variables and the output. We have $n = 500$ observations (one for each firm) on $p = 3$ variables (profit, number of employees, industry).
- b) This is a classification problem, and we want to predict the outcome variable. We have $n = 20$ (20 similar products previously launched) observations on $p = 13$ variables (price charged, marketing budget, competition price, and 10 other variables).
- c) This is a regression problem, as we want to predict the outcome variable. We have $n = 52$ (one for each week in 2012) observations on $p = 3$ (% change in US market, % change in British market, % change in German market) variables.

Exercise 5.3. We will now think of some real-life applications for different statistical learning methods.

- a) Describe three real-life applications in which classification might be useful. Describe the response, as well as the predictors. Is the goal of each application inference (describing the relationship) or prediction? Explain your answer.
- b) Describe three real-life applications in which regression might be useful. Describe the response, as well as the predictors. Is the goal of each application inference (describing the relationship) or prediction? Explain your answer.

Solution:

- a)
 - Email Spam Detection

Application: Filtering spam emails from legitimate emails.

Response (Target Variable): Binary indicator where 1 represents spam and 0 represents non-spam (legitimate) emails.

Predictors:

- Presence of certain keywords (e.g., “win”, “free”, “limited time”).
- Sender’s email address or domain.
- Frequency of certain characters or phrases.
- Email metadata (e.g., time sent, subject line length).
- Email body length.
- Attachment presence and type.

Goal: Prediction

Explanation: The primary goal is to predict whether a new, unseen email is spam or not. The focus is on accurately classifying future emails to improve user experience and security by filtering out unwanted or malicious content.

- Medical Diagnosis

Application: Diagnosing diseases based on patient data.

Response: Disease status, which can be binary (e.g., presence or absence of a disease) or multiclass (e.g., type of disease such as diabetes, heart disease, etc.).

Predictors:

- Demographic information (e.g., age, gender).
- Clinical measurements (e.g., blood pressure, cholesterol levels, glucose levels, resting heart rate).
- Medical history (e.g., family history of disease, previous diagnoses).
- Lifestyle factors (e.g., smoking status, physical activity).
- Results from diagnostic tests (e.g., imaging results, blood tests).

Goal: Prediction

Explanation: The main objective is to predict the disease status of new patients based on their medical data. Accurate prediction can lead to early diagnosis and treatment, which can significantly improve patient outcomes.

- Customer Churn Prediction

Application: Identifying customers likely to stop using a service or subscription (churn).

Response: Binary indicator where 1 represents a customer who will churn and 0 represents a customer who will stay.

Predictors:

- Customer demographics (e.g., age, income).
- Usage patterns (e.g., frequency of use, duration of service usage).
- Customer service interactions (e.g., number of support tickets, call center interactions).

- Transaction history (e.g., recent purchases, payment history).
- Subscription details (e.g., plan type, tenure with the company).
- Engagement metrics (e.g., login frequency, feature usage).

Goal: Prediction

Explanation: The goal is to predict which customers are at risk of churning so that proactive measures can be taken to retain them. This might include targeted marketing campaigns, personalized offers, or improvements in customer service. Accurate predictions can help companies reduce churn rates and maintain a stable customer base.

b) • House Price Prediction

Application: Estimating the selling price of residential properties. **Response:** Sale price of the house.

Predictors (Features):

- Size of the house (square footage).
- Number of bedrooms and bathrooms.
- Location (e.g., neighborhood, proximity to schools and amenities).
- Age of the property.
- Lot size.
- Presence of features such as a garage, swimming pool, or garden.

Goal: Prediction

Explanation: The primary objective is to predict the price of a house based on its features. This is useful for real estate agents, buyers, and sellers to make informed decisions. Accurate predictions can help in setting competitive prices and understanding market trends.

• Advertising Effectiveness

Application: Measuring the impact of advertising spend on sales revenue.

Response: Sales revenue.

Predictors (Features):

- Amount spent on different advertising channels (e.g., TV, radio, online).
- Timing and duration of advertising campaigns.
- Type of advertisements (e.g., promotional, informational).
- Market conditions (e.g., economic indicators, competition activity).
- Demographic information of the target audience.

Goal: Inference

Explanation: The main goal is to understand the relationship between advertising spend and sales revenue. This involves estimating the return on investment for different advertising channels and identifying the most effective strategies. By understanding these relationships, companies can optimize their advertising budgets and strategies.

• Energy Consumption Modeling

Application: Estimating energy consumption in buildings based on various factors.

Response: Energy consumption (e.g., kilowatt-hours).

Predictors:

- Building size (e.g., square footage).
- Number of occupants.
- Building age.
- Type of heating/cooling systems.
- Weather conditions (e.g., temperature, humidity).
- Usage patterns (e.g., hours of operation).

Goal: Prediction and Inference

Explanation: This application has a dual goal. The prediction aspect focuses on estimating future energy consumption to manage resources, plan for peak demand, and optimize

energy usage. The inference aspect aims to understand how different factors, such as building characteristics and weather conditions, affect energy consumption. This can help in designing energy-efficient buildings and implementing effective energy-saving measures.

Exercise 5.4. The following training data, containing six observations on three predictors and one qualitative response variable, is given.

```
train
#   x1 x2 x3    y
# 1  0  3  0   red
# 2  2  0  0   red
# 3  0  1  3   red
# 4  0  1  2 green
# 5 -1  0  1 green
# 6  1  1  1   red
```

Suppose we wish to use this dataset to predict a value for the response Y given the new observation:

```
new
#   x1 x2 x3
# 1  0  0  0
```

using k-nearest neighbors.

- Compute the euclidean distance between each observation and the new test point, **new**.
- What is your prediction for $k = 1$? Justify your answer.
- What is our prediction for $k = 3$? Justify your answer.

Solution:

- The euclidean distances are the following:

```
library(dplyr)
train |>
  mutate(dist = sqrt((x1 - 0)^2 + (x2 - 0)^2 + (x3 - 0)^2)) |>
  arrange(dist)
#   x1 x2 x3    y    dist
# 1 -1  0  1 green 1.414214
# 2  1  1  1   red 1.732051
# 3  2  0  0   red 2.000000
# 4  0  1  2 green 2.236068
# 5  0  3  0   red 3.000000
# 6  0  1  3   red 3.162278
```

- For $k = 1$, the prediction will be **green** since the nearest neighbor of $(0,0,0)$ is the observation $(-1,0,1)$.
- For $k = 3$, the prediction will be **red** since the three nearest neighbors of $(0,0,0)$ are $(-1,0,1)$, $(1,1,1)$, and $(2,0,0)$ where **red** wins the majority vote with two occurrences.

Exercise 5.5. Data are provided on the shoulder girth and height of a group of individuals. The mean shoulder girth is 107.20 cm (standard deviation: 10.37 cm), and the mean height is 171.14 cm (standard deviation: 9.41 cm). The correlation coefficient between height and shoulder girth is 0.67.

- Write the equation of the fitted regression model for predicting height.
- Interpret the slope and the intercept in this context.
- Calculate R^2 of the regression line for predicting height from shoulder girth, and interpret it in the context of the application.
- A randomly selected student from your class has a shoulder girth of 100 cm. Predict the student's height using the model.
- The student from d) is 160 cm tall. Calculate the residual and explain what this residual means.
- A one-year-old has a shoulder girth of 56 cm. Would it be appropriate to use this linear model to predict the height of this child?

Solution:

- First, the slope is calculated.

$$\hat{\beta}_1 = \frac{s_y}{s_x} \cdot r_{(x,y),n} = \frac{9.41}{10.37} \cdot 0.67 \approx 0.61$$

Next, make use of the fact that the regression line passes through the point $(\bar{x}, \bar{y})$.

$$\begin{aligned}\bar{y} &= \hat{\beta}_0 + \hat{\beta}_1 \bar{x} \\ 171.14 &= \hat{\beta}_0 + 0.61 \cdot 107.20 \\ \hat{\beta}_0 &= 171.14 - 0.61 \cdot 107.20 \approx 105.75\end{aligned}$$

The fitted regression line is given by

$$\hat{y} = 105.75 + 0.61 \cdot x,$$

where y represents height and x denotes shoulder girth.

- $\hat{\beta}_1$: For each additional centimeter of shoulder girth, the model predicts an average increase in height of 0.61 centimeters.
 $\hat{\beta}_0$: The model predicts that individuals with a shoulder girth of 0 cm would have an average height of 105.75 cm. However, the intercept is not meaningful in this context and serves only to position the regression line.
- The R squared value is equal to $R^2 = r_{(x,y),n}^2 = 0.67^2 = 0.4489$.
The model accounts for approximately 44% of the variation in heights.
- To predict height from shoulder girth, substitute $x = 100$ into the regression equation:

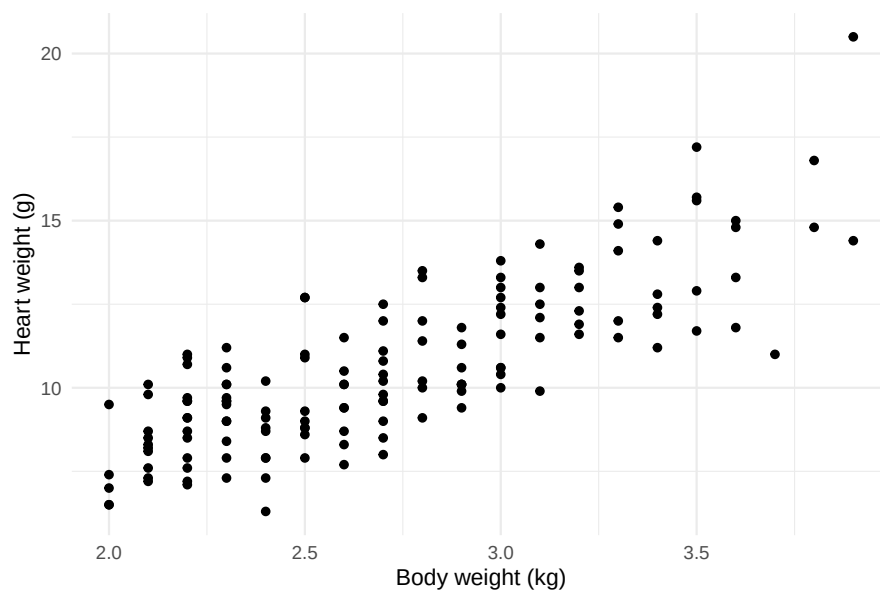
$$\hat{y} = 105.14 + 0.61 \cdot x = 105.75 + 0.61 \cdot 100 = 166.75 \text{ cm}$$

- The residual is calculated as $e_i = y_i - \hat{y}_i = 160 - 166.75 = -6.75$ cm. A negative residual indicates that the model overestimated the student's height.
- No. Predicting the height of a child with a shoulder girth of 56 cm would require extrapolation, as the model was developed using data from individuals with shoulder girths between approximately 90 and 130 cm. The linear relationship may not be valid outside this range.

Exercise 5.6. The following regression model

```
mod <- lm(Hwt ~ Bwt, data = cats)
```

is used for predicting the heart weight **Hwt** (in g) of cats from their body weight **Bwt** (in kg).



The coefficients are estimated using a dataset of 144 domestic cats. One obtains the following least squares estimates

```
coef(mod)
# (Intercept)      Bwt
#      -0.357      4.034
```

- Write out the fitted linear model.
- Interpret the intercept.
- Interpret the slope.
- Identify the R^2 value in

```
broom::glance(mod)
# # A tibble: 1 x 12
#   r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC    BIC
#   <dbl>      <dbl> <dbl>     <dbl>    <dbl> <dbl> <dbl> <dbl> <dbl>
# 1     0.647        0.644  1.45      260. 6.97e-34     1  -257.  520.  529.
# # i 3 more variables: deviance <dbl>, df.residual <int>, nobs <int>
```

and give an interpretation of the computed R^2 value.

- Calculate the empirical correlation coefficient between the response and the predictor variable.

Solution: a) The fitted model is given by

$$\hat{y} = -0.357 + 4.034 \cdot x,$$

where y denotes the heart and x the body weight.

- b) The heart weight of cats with zero body weight is expected to be -0.357 on average. This is obviously not a meaningful value; it just adjusts the height of the regression line.
- c) For each additional kilogram in cats' weights, we expect their hearts to be heavier by 4.034 grams, on average.
- d) The R^2 value is the first entry in the output of `glance()` and hence equal to 0.64662. The model using body weight as a predictor can explain around 64.66% of the variability in the weights of cats' hearts.
- e) Since the correlation is equal to the square root of the R^2 , we get

$$r_{(x,y),n} = \sqrt{0.64662} = 0.804127.$$

Exercise 5.7. In this exercise, we are using a forward-selection algorithm, where we attempt to add variables to the model one at a time. The table below displays the initial step of the forward-selection process, beginning with an intercept model and considering the variables `ethnicity`, `gender`, and `cls_students` within the scope. Based on this table, which variable should be added to the model in the first step, and why?

```
Start:  AIC=-562.99
score ~ 1
```

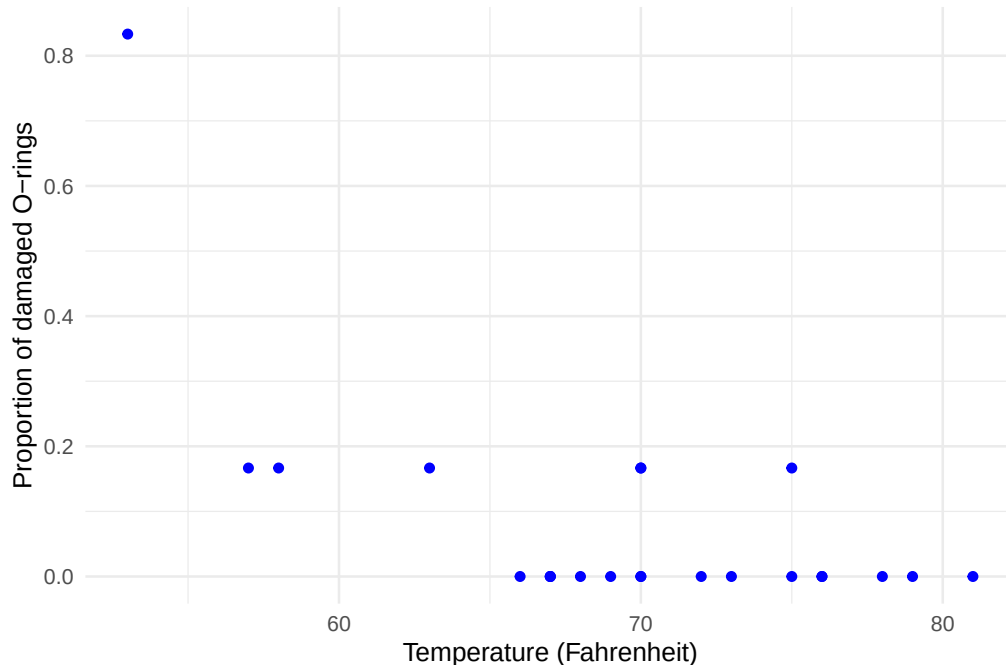
	Df	Sum of Sq	RSS	AIC
+ cls_students	1	0.09216	136.56	-561.30
+ ethnicity	1	0.78567	135.87	-563.66
+ gender	1	2.26021	134.39	-568.71
<none>			136.65	-562.99

Hint: The table is not given in standard form.

Solution:

The table contains RSS and AIC. Both should be as small as possible. When we look at the table, we notice that it is not given in standard form, meaning it is not ordered increasingly with respect to the AIC values. Therefore, we need to determine which action results in the smallest AIC value, and it turns out to be adding `gender` to the model.

Exercise 5.8. On January 28, 1986, the Challenger space shuttle was scheduled for a routine launch. Seventy-three seconds after liftoff, the shuttle disintegrated, causing the deaths of all seven crew members. The investigation identified a critical seal, known as an O-ring, as a primary factor in the disaster. Analysis revealed a strong association between ambient temperature at launch and O-ring damage. The following plot displays the proportion of damaged O-rings at each recorded temperature. Data on O-ring performance were collected from 23 shuttle missions, with ambient temperature recorded for each launch. Six O-rings were tested per mission.



- a) The logistic model fit to these data may be written as

$$\log \left(\frac{\hat{p}(x)}{1 - \hat{p}(x)} \right) = 11.6630 - 0.2162 \times x$$

where $\hat{p}$ is the model-estimated probability that an O-ring will become damaged and x is the temperature in Fahrenheit. Use the model to calculate the probability of O-ring damage at ambient temperatures of 51, 53, and 55 degrees Fahrenheit. The model-estimated probabilities for various additional ambient temperatures are provided below.

$$\begin{array}{llll} \hat{p}(57) = 0.341 & \hat{p}(59) = 0.251 & \hat{p}(61) = 0.179 & \hat{p}(63) = 0.124 \\ \hat{p}(65) = 0.084 & \hat{p}(67) = 0.056 & \hat{p}(69) = 0.037 & \hat{p}(71) = 0.024 \end{array}$$

- b) Add the model-estimated probabilities from a) on the plot, then connect these dots using a smooth curve to represent the model-estimated probabilities.
Hint: If you want to add the curve using R, you can do so. The dataset is called `openintro::orings`.
- c) Describe any concerns you may have regarding applying logistic regression in this application, and note any assumptions that are required to accept the model's validity.

Solution:

a) The calculation of the probabilities are as follows:

$$\begin{aligned}\log\left(\frac{\hat{p}_{51}}{1-\hat{p}_{51}}\right) &= 11.6630 - 0.2162 \cdot 51 \rightarrow \hat{p}_{51} = \frac{e^{11.6630-0.2162 \cdot 51}}{1 + e^{11.6630-0.2162 \cdot 51}} = 0.654 \\ \log\left(\frac{\hat{p}_{53}}{1-\hat{p}_{53}}\right) &= 11.6630 - 0.2162 \cdot 53 \rightarrow \hat{p}_{53} = \frac{e^{11.6630-0.2162 \cdot 53}}{1 + e^{11.6630-0.2162 \cdot 53}} = 0.551 \\ \log\left(\frac{\hat{p}_{55}}{1-\hat{p}_{55}}\right) &= 11.6630 - 0.2162 \cdot 55 \rightarrow \hat{p}_{55} = \frac{e^{11.6630-0.2162 \cdot 55}}{1 + e^{11.6630-0.2162 \cdot 55}} = 0.443\end{aligned}$$

b) Let's start by recreating the plot from the problem statement

```
library(openintro)
library(tidyverse)

# we need to compute the number of damaged O-rings per temperature.

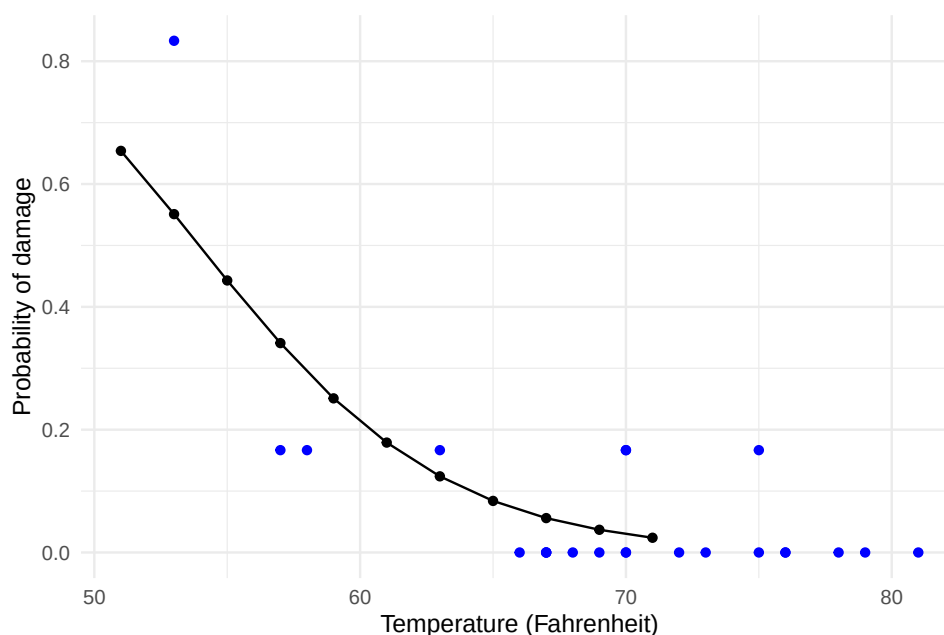
temp_prob <- orings |>
  mutate(prob = damaged / 6)
```

The calculated probabilities are

```
temp_mod_prob <- tibble(
  temperature = seq(51, 71, by = 2),
  prob = c(0.654, 0.551, 0.443, 0.341, 0.251, 0.179, 0.124,
           0.084, 0.056, 0.037, 0.024))
```

Now we can add the predicted probabilities to the plot and connect them with a smooth line.

```
ggplot(temp_prob, aes(x = temperature, y = prob)) +
  geom_point(colour = "blue") +
  labs(x = "Temperature (Fahrenheit)", y = "Probability of damage") +
  geom_point(data = temp_mod_prob) +
  geom_line(data = temp_mod_prob) +
  theme_minimal()
```



- c) In the logistic regression model, we assume that observations are independent of each other (the same is true for the linear regression model). In this case, each O-ring must be independent of the others. This assumption might be critical. For example, if the O-ring manufacturing process has changed dramatically over the time of the different shuttle launches, then we should be skeptical of independence and therefore also of the model.

Exercise 5.9. Determine which of the following statements are true and which are false. For each false statement, provide an explanation for why it is incorrect.

- a) Suppose we have the first two observations from a logistic regression model. In observation 1, the first variable takes the value $x_1 = 6$, and in observation 2, it takes the value $x_1 = 4$. Suppose errors were made in recording the two observations. The correct value for the first observation is $x_1 = 7$ (instead of 6), and for the second observation, $x_1 = 5$ (instead of 4). After correcting these values, the predicted probability from the logistic regression model would increase by the same amount for each observation.
- b) When using a logistic regression model, the model cannot predict a probability that is negative or greater than 1.
- c) Because logistic regression predicts probabilities of outcomes, the observations used to build a logistic regression model need not be independent.
- d) When we fit a logistic regression model, we usually perform model selection using adjusted R^2 .

Solution:

- a) False: The log-odds change by the same amount, but these changes are not generally proportional to a change in probability.
- b) True: By definition, the logistic function yields probabilities $p_i = \frac{e^{\eta_i}}{1+e^{\eta_i}}$, which are strictly positive and bounded above by one.
- c) False: Standard logistic regression requires independent observations. When this assumption is violated, specialized methods must be employed.
- d) False: Model selection in logistic regression is typically performed using the AIC, not adjusted R^2 .

6 Foundations of Inference

Exercise 6.1. A recent survey of 840 adults in Germany found that 268 respondents said they would be unable to cover an unexpected 5000-Euro expense without borrowing money or going into debt.

- a) What population is under consideration in the dataset?
- b) What parameter is being estimated?
- c) What is the point estimate for the parameter?
- d) Use the function `rep()` and `data.frame()` to create the sample data in R.
Hint: Create a data frame with one column named `expense`, containing 840 entries. 572 entries should be `"cover"` and the remaining 268 entries `"not cover"`.
- e) Use functions from the `infer` package to create a bootstrap distribution for the point estimate from part c) based on 1000 resamples.
- f) Compute the standard deviation of the point estimator based on the 1000 point estimates computed in part e).
- g) A politician claims that 40% of German adults cannot cover a 5000-Euro emergency expense. Based on the bootstrap distribution, should she be surprised by the data?

Exercise 6.2. A national poll conducted in 2013 found that 52% of U.S. adult Twitter users get at least some news on Twitter, and the standard error for this estimate was 2.4%. Identify each of the following statements as true or false. Provide an explanation to justify each of your answers.

- a) The data provide evidence with a confidence level of $1 - \alpha = 0.99$, indicating that more than half of U.S. adult Twitter users receive some news through Twitter.

Remark: This is a conservative statement.

- b) Since the standard error is 2.4%, we can conclude that 97.6% of all U.S. adult Twitter users were included in the study.
- c) If we want to reduce the standard error of the estimate, we should collect less data.
- d) If we construct a 90% confidence interval for the percentage of U.S. adults Twitter users who get some news through Twitter, this confidence interval will be wider than a corresponding 99% confidence interval.

Exercise 6.3. We are provided with a 95% confidence interval for the mean waiting time at an emergency room (ER) of (128, 147) minutes. Answer the following questions based on this interval.

- a) A local newspaper claims that the average waiting time at this ER exceeds 3 hours. Is this claim supported by the confidence interval? Explain your reasoning.
- b) The Dean of Medicine at this hospital claims the average wait time is 2.2 hours. Is this claim supported by the confidence interval? Explain your reasoning.
- c) Without actually calculating the interval, determine if the claim of the Dean from b) would be supported based on a 99% confidence interval?

Exercise 6.4. State the null and alternative hypotheses in both words and symbols for each scenario described below.

- a) In Florence, Italy, a new regulation requires gelaterie (ice-cream shops) to display the calorie content of every scoop of gelato.

Before this rule, the average caloric intake per customer at a popular gelato shop was 950 calories.

After calorie posting began, a nutrition researcher collected a random sample of customers to see whether calorie labeling changed how much people eat.

She believes there is a difference in the average number of calories consumed.

- b) The government of the Faroe Islands wants to understand last year's alcohol consumption among adults.

The Nordic regional average is that 76% of adults consumed alcohol at least once in the past year.

To investigate whether the Faroese population differs from this regional value, officials surveyed a random sample of 620 adults regarding their alcohol consumption.

Exercise 6.5. A study suggests that 25% of 25-year-olds are married. However, you believe this is incorrect and decide to collect your own sample for a hypothesis test. In a random sample of 25-year-olds from census data, consisting of 776 individuals, it was found that 24% of them are married. A friend of yours offered to help you set up the hypothesis test and came up with the following hypotheses. Indicate any errors you see.

$$H_0 : \hat{\theta} = 0.24 \quad \text{vs.} \quad H_A : \hat{\theta} \neq 0.24.$$

Exercise 6.6. A chocolate producer suspects irregularities in her production. For quality control, she randomly selects 15 bars and measures their weight in grams. The variable $\mathbf{x}$ contains the sample observations. The producer has identified an issue with the average weight, which should be 100g. You can assume that the weight follows a normal distribution, meaning that $\mathbf{x}$ contains realizations of 15 i.i.d. random variables, denoted as $X_1, \dots, X_{15}$. Each X_i follows a normal distribution with mean μ and variance σ^2 , with both being unknown.

- Formulate null hypothesis and alternative hypothesis for a two-sided test for the average weight μ .
- Test the null hypothesis from part a) at the significance level $\alpha = 0.05$. Justify your answer using the hint provided.

Hint:

```
c(sum(x), sd(x))
```

```
# [1] 1482.07    1.69
```

```
pt(seq(-3, -2, by = 0.1), df = 14)
```

```
# [1] 0.00477576 0.00582103 0.00709016 0.00862833 0.01048867 0.01273333
```

```
# [7] 0.01543446 0.01867514 0.02255015 0.02716657 0.03264398
```

Exercise 6.7. Re-examine the data set provided in the file `gifted.csv`, available on Moodle.

Remember: Researchers collected a simple random sample of 36 children identified as gifted in a large city. The variables included in the data set are as follows:

- **score** Score in test of analytical skills.
- **fatheriq** Father's IQ.
- **motheriq** Mother's IQ.
- **speak** Age in months when the child first said 'mummy' or 'daddy'.
- **count** Age in months when the child first counted to 10 successfully.
- **read** Average number of hours per week the child's mother or father reads to the child.
- **edutv** Average number of hours per week the child watched an educational program on TV during the past three months.
- **cartoons** Average number of hours per week the child watched cartoons on TV during the past three months.

The researchers believe that gifted children spend more time watching cartoons than educational programs.

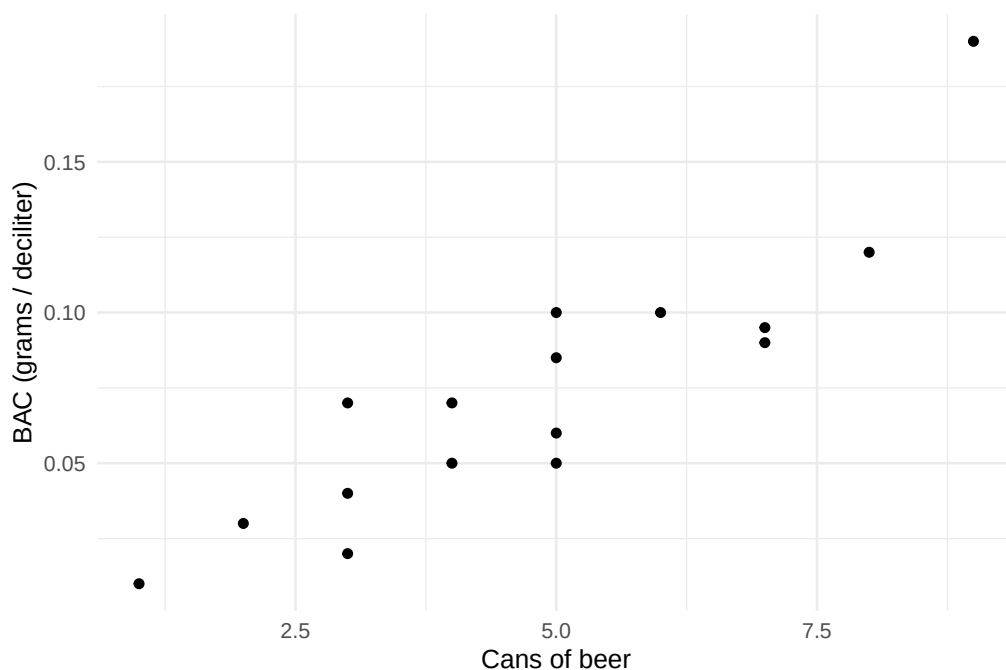
- a) State the null and alternative hypotheses in order to test the researchers' assumption.
- b) Use the simulation-based approach for testing the null hypothesis from part a). Decide if you can reject the null hypothesis at a significance level of $\alpha = 0.05$. State your conclusion in the context of the data.

7 Inference for linear regression

Exercise 7.1. Many people believe that gender, weight, drinking habits, and many other factors are much more important in predicting blood alcohol content (BAC) than simply considering the number of drinks a person consumed. Here, we examine data from 16 student volunteers recruited at a German university who each drank a randomly assigned number of cans of beer. These students were evenly divided between men and women, and they differed in weight and drinking habits. Thirty minutes later, a police officer measured their blood alcohol content (BAC) in grams of alcohol per deciliter of blood.

The following scatterplot visualizes the observed data and the R output summarizes the results of the fitted regression model.

```
# # A tibble: 2 x 5
#   term      estimate std.error statistic    p.value
#   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
# 1 (Intercept) -0.0127  0.0126   -1.00  0.332
# 2 beers       0.0180  0.00240    7.48 0.00000297
```



- Describe the relationship between the number of cans of beer and BAC.
- Write the equation of the fitted regression line. Interpret the slope and intercept in the context of the data.
- Do the data provide strong evidence that drinking more cans of beer is associated with an increase in blood alcohol? State the null and alternative hypotheses, report the p-value, and state your conclusion.
- The correlation coefficient for the number of cans of beer and BAC is 0.89. Calculate R^2 and interpret it in context.
- Suppose we visit a bar, ask people how many drinks they have had, and also take their BAC. Do you think the relationship between the number of drinks and BAC would be as strong as the relationship found in the study?

Exercise 7.2. We are given the regression output from a model for predicting the heart weight (in g) of cats from their body weight (in kg). The coefficients are estimated using a dataset of 144 domestic cats. The model output is also provided below.

```
# # A tibble: 2 x 5
#   term      estimate std.error statistic    p.value
#   <chr>      <dbl>    <dbl>    <dbl>    <dbl>
# 1 (Intercept) -0.3567    0.6923   -0.5152 6.072e- 1
# 2 Bwt         4.034     0.2503   16.12   6.969e-34
```

- We see that the point estimate for the slope is positive. What are the hypotheses for evaluating whether body weight is positively associated with heart weight in the population of domestic cats?
- Test the hypotheses from part a) using a 5% significance level, and then report the conclusion of the test in the context of the data.
- Calculate a 95% confidence interval for the slope of body weight, and interpret it in the context of the data.

Hint:

```
qt(c(0.9, 0.95, 0.975), df = 142)
# [1] 1.29 1.66 1.98
```

- Do your results from the hypothesis test and the confidence interval agree? Explain.

Exercise 7.3. The data set `Cars93` contains 93 observations on motor cars sold in the US in 1993. To explain the logarithm of the variable `MPG.city` (City MPG - miles per US gallon by EPA rating), a linear model was fitted using the explanatory variables

- **EngineSize:** Engine size (liters)
- **Weight:** Weight (pounds)
- **Passengers:** Passenger capacity (persons)
- **Price:** Mid-range Price (in \$1000): average of `Min.Price` and `Max.Price`.

The summary of this model contains some missing values (NA).

```
summary(model)

#
# Call:
# lm(formula = log(MPG.city) ~ ., data = cars)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -0.2631 -0.0619  0.0058  0.0460  0.4169
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  4.06e+00   7.24e-02     NA      NA
# EngineSize  -3.80e-03   2.03e-02     NA      NA
# Weight       -3.14e-04   4.63e-05     NA      NA
# Passengers   6.31e-03   1.47e-02     NA      NA
# Price        -2.03e-03   1.70e-03     NA      NA
#
# Residual standard error: 0.106 on 88 degrees of freedom
# Multiple R-squared:  0.787,    Adjusted R-squared:  0.778
# F-statistic: 81.4 on 4 and 88 DF,  p-value: <2e-16
```

- Identify all variables, which are significant at the level $\alpha = 0.05$. Justify your answer.
- Give an interpretation of the slope of all variables that have a significant effect due to a).
- Now a second model was fitted:

```
model_small <- lm(log(MPG.city) ~ Weight + Price, data = Cars93)
```

which has an adjusted R^2 of 0.781858. Which model would you prefer based on this information? Justify your answer.

Hint:

```
qt(0.025, df = 88, lower.tail = FALSE)
# [1] 1.98729
```

Exercise 7.4. In 41 US cities, sulphur dioxide concentrations in the air were analysed with respect to several climatic and geographical variables. A multiple linear regression model was fitted using the three explanatory variables `temp` (average annual temperature in degrees F), `man_fac` (number of manufacturers employing >20 workers), and `wind` (average annual wind speed in miles per hour). The dependent variable was the logarithmic annual average sulphur dioxide concentration in micrograms per square meter `log(sd_qm)`.

The following summary (with some missing values) was obtained.

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	7.4892948	0.9927641	7.544	5.41e-09	***
temp	-0.0556772	0.0116911	-4.762	2.93e-05	***
num_manfac	0.0005633	0.0001446	3.894	0.000398	***
wind	-0.1580268	0.0597864	-2.643	0.011972	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4972 on 37 degrees of freedom

Multiple R-squared: 0.5364, Adjusted R-squared: NA

F-statistic: 14.27 on 3 and 37 DF, p-value: 2.473e-06

- State the equation of the fitted regression model.
- Formulate the null and alternative hypotheses to test if `wind` has a significant effect on the sulphur dioxide concentration. Decide if it has a significant effect at level $\alpha = 0.05$.
- Compute the adjusted R^2 value.
- Why did it make sense to consider the logarithmic pollutant concentrations?

Hint:

```
var(log(sd_qm))
# [1] 0.493223
```